

SDM-NET: Deep Generative Network for Structured Deformable Mesh

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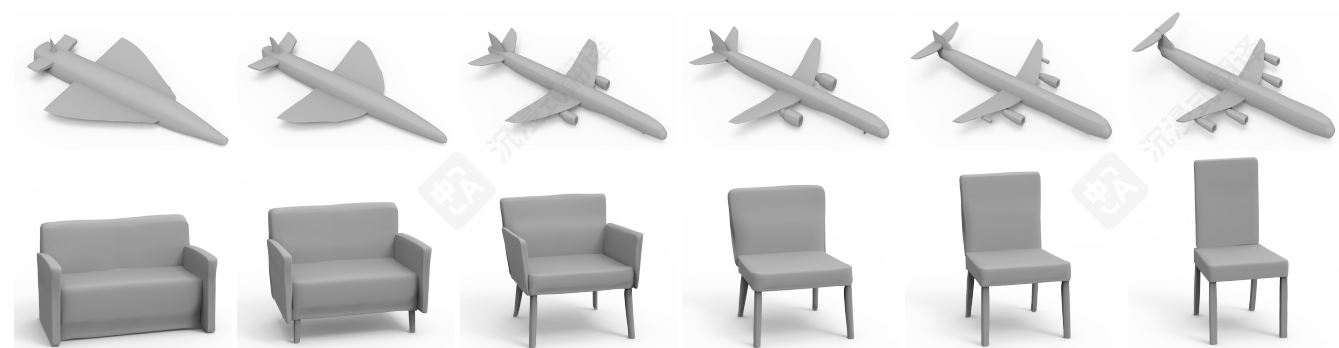


Fig. 1. Our deep generative neural network, SDM-NET, produces structured meshes composed of deformable parts. Part structures and geometries are jointly encoded into a latent space by an autoencoder, enabling quality 3D shape generation. We show shape interpolation results exhibiting flexible structure and fine geometric details. This is achieved by linearly interpolating airplane and chair latent codes and then reconstruction from the in-between codes.

We introduce SDM-NET, a deep generative neural network which produces *structured deformable meshes*. Specifically, the network is trained to generate a spatial arrangement of closed, deformable mesh parts, which respects the global part structure of a shape collection, e.g., chairs, airplanes, etc. Our key observation is that while the overall structure of a 3D shape can be complex, the shape can usually be decomposed into a set of parts, each homeomorphic to a box, and the finer-scale geometry of the part can be recovered by *deforming* the box. The architecture of SDM-NET is that of a *two-level variational autoencoder* (VAE). At the part level, a PartVAE learns a deformable model of part geometries. At the structural level, we train a Structured Parts VAE (SP-VAE), which *jointly* learns the part structure of a shape collection and the part geometries, ensuring the coherence between global shape structure and surface details. Through extensive experiments and comparisons with the state-of-the-art deep generative models of shapes, we demonstrate the superiority of SDM-NET in generating meshes with visual quality, flexible topology, and meaningful structures, benefiting shape interpolation and other subsequent modeling tasks.

CCS Concepts: • **Computing methodologies** → **Shape modeling**.

Additional Key Words and Phrases: Shape representation, variational autoencoder, structure, deformation, geometric details, generation, interpolation

1 INTRODUCTION

Triangle meshes have been the dominant 3D shape representation in computer graphics, for modeling, rendering, manipulation, and animation. However, as deep learning becomes pervasive in visual

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computing, most deep convolutional neural networks (CNNs) developed for shape modeling and analysis have resorted to other representations including voxel grids [Girdhar et al. 2016; Qi et al. 2016; Wu et al. 2016, 2015], shape images [Sinha et al. 2016; Su et al. 2015], and point clouds [Qi et al. 2017; Yin et al. 2018]. One of the main reasons is that the non-uniformity and irregularity of triangle tessellations do not naturally support conventional convolution and pooling operations [Hanocka et al. 2019]. Yet, advantages of meshes over other shape representations should not be overlooked.

Compared to voxels, meshes are more compact and better suited to representing finer surface details. Compared to points, meshes are more controllable and exhibit better visual quality. There have been recent attempts at developing mesh-specific convolutional operators designed for triangle tessellations [Hanocka et al. 2019; Poulenard and Ovsjanikov 2018]. Current deep generative models for meshes are limited to either genus-zero meshes [Hamu et al. 2018; Maron et al. 2017] or meshes sharing the same connectivity [Gao et al. 2018; Tan et al. 2018]. Patch-based models which cover a shape with planar [Wang et al. 2018b] or curved [Groueix et al. 2018] patches, are more adaptive, but surface quality is often tampered by visible seams and the patches are otherwise unstructured and incoherent.

In this paper, we introduce a novel deep generative neural network for meshes which overcomes the above limitations. Our key observation is that while the overall structure of a 3D shape can be complex, the shape can usually be decomposed into a set of *parts*, each homeomorphic to a box, and the finer-scale geometry of the part can be recovered by *deforming* the box. Hence, the

SDM-NET: 用于结构化可变形网格的深度生成网络

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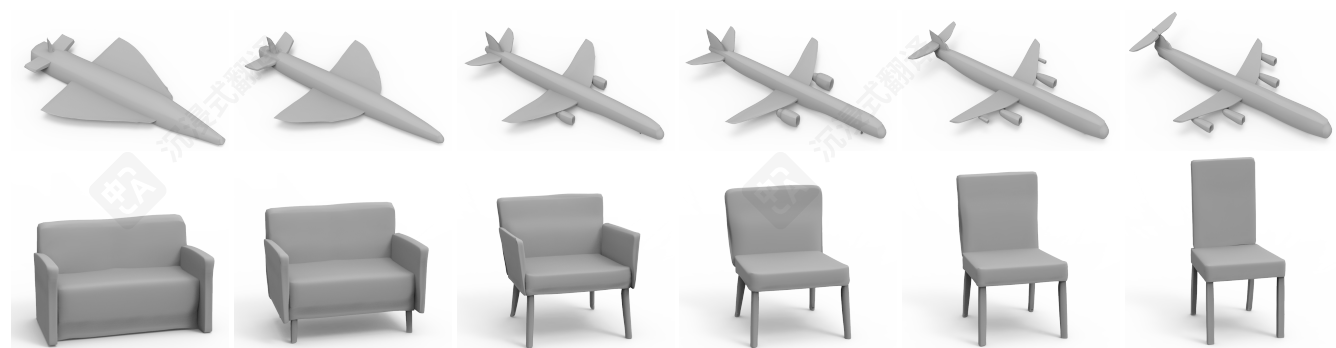


图1. 我们的深度生成神经网络SDM-NET生成由可变形部分组成的结构化网格。部分结构和几何形状被自动编码器联合编码到潜在空间中, 从而实现高质量的3D形状生成。我们展示了形状插值结果, 展示了灵活的结构和精细的几何细节。这是通过线性插值飞机和椅子的潜在代码, 然后从中间代码进行重建来实现的。

我们介绍了SDM-NET, 一种深度生成神经网络, 用于生成结构化可变形网格。具体来说, 该网络被训练以生成一组闭合的可变形网格部件的空间排列, 这些排列尊重形状集合 (例如椅子、飞机等) 的整体部件结构。我们的关键观察是, 虽然3D形状的整体结构可能很复杂, 但通常可以分解为一组部件, 每个部件同胚于一个盒子, 并且可以通过变形盒子来恢复部件的精细尺度几何。SDM-NET的架构是两级变分自编码器 (VAE)。在部件级别, PartVAE学习部件几何的可变形模型。在结构级别, 我们训练了一个结构化部件VAE (SP-VAE), 它联合学习形状集合的部件结构和部件几何, 确保全局形状结构与表面细节之间的协调。通过广泛的实验并与最先进的形状深度生成模型进行比较, 我们证明了SDM-NET在生成具有视觉质量、灵活拓扑和有意义结构的网格方面的优越性, 这有助于形状插值和其他后续建模任务。

CCS Concepts: • **Computing methodologies** → **Shape modeling**.

其他关键词和短语: 形状表示、变分自编码器、结构、变形、几何细节、生成、插值

1 引言

三角形网格一直是计算机图形学中三维形状表示的主流, 用于建模、渲染、操作和动画。然而, 随着深度学习在视觉领域变得越来越普遍

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随着深度学习在视觉计算中变得普及, 大多数用于形状建模和分析的深度卷积神经网络 (CNN) 已经转向其他表示方法, 包括体素网格 [Girdhar等人, 2016年; Qi等人, 2016年; Wu等人 2016, 2015], 形状图像 [Sinha等人, 2016年; Su等人.2015], 以及点云 [Qi等人, 2017年; Yin等人 2018]。主要原因之一是三角形镶嵌的非均匀性和不规则性, 无法自然支持传统的卷积和池化操作 [Hanocka等人 2019]。然而, 网格相对于其他形状表示方法的优势不应被忽视。

与体素相比, 网格更紧凑, 更适合表示更精细的表面细节。与点相比, 网格更易于控制, 并表现出更好的视觉效果。最近已有尝试开发针对三角形剖分的网格特定卷积算子 [Hanocka 等人 2019; Poulenard 和 Ovsjanikov 2018]。当前的网格深度生成模型仅限于零亏格网格 [Hamu 等人 2018; Maron 等人 2017] 或具有相同连接性的网格 [Gao 等人 2018; Tan 等人 2018]。基于补丁的模型 (用平面 [Wang 等人 2018b] 或曲面 [Groueix 等人 2018] 补丁覆盖形状), 更具适应性, 但表面质量常因可见接缝而受损, 且补丁通常无结构且不连贯。

在本文中, 我们介绍了一种用于网格的新型深度生成神经网络, 该网络克服了上述限制。我们的关键观察是, 虽然三维形状的整体结构可能很复杂, 但形状通常可以分解为一组部分, 每个部分同胚于一个盒子, 并且可以通过变形盒子来恢复该部分的精细尺度几何。因此,

architecture of our network is that of a *two-level* variational autoencoder (VAE) [Kingma and Welling 2013] which produces *structured deformable meshes* (SDM). At the part level, a PartVAE learns a deformable model of shape parts, by means of autoencoding fixed-connectivity, genus-zero meshes. At the structural level, we train a Structured Parts VAE (SP-VAE), which *jointly* learns the part structure of a shape collection and the part geometries, ensuring the coherence between global shape structure and surface details.

We call our network SDM-NET, as it is trained to generate structured deformable meshes, that is, a spatial arrangement of closed, deformable mesh parts, which respects the global part structure (e.g., symmetry and support relations among shape parts) of a shape collection, e.g., chairs, airplanes, etc. However, our network can generate shapes with a varying number of parts, up to a maximum count. Besides the advantages afforded by meshes mentioned above, a structured representation allows shapes generated by SDM-NET to be immediately reusable, e.g., for assembly-based modeling [Mitra et al. 2013]. In addition, the deformability of the mesh parts further facilitates editing and interpolation of the generated shapes.

SDM-NET is trained with a shape collection equipped with a consistent part structure, e.g., semantic segmentation. However, the shapes in the collection can be with arbitrary topologies and mesh connectivities. Such data sets are now widely available, e.g., ShapeNet [Chang et al. 2015] and PartNet [Mo et al. 2019b], to name a few. While direct outputs from SDM-NET are not watertight meshes, each part is.

In summary, the main contributions of our work are:

- The first deep generative neural network which produces structured deformable meshes.
- A novel network architecture corresponding to a two-level variational autoencoder which jointly learns shape structure and geometry. This is in contrast to the recent work, GRASS [Li et al. 2017], which learns shape structure and part geometry using separate networks.
- A support-based part connection optimization to ensure the generation of plausible and physically valid shapes.

Figure 1 demonstrates the capability of our SDM-NET to reconstruct shapes with flexible structure and fine geometric details. By interpolating in the latent space, new plausible shapes with substantial structure change are generated.

Through extensive experiments and comparisons with the state-of-the-art deep generative models of shapes, we demonstrate the superiority of SDM-NET in generating quality meshes and shape interpolations. We also show the structured deformation meshes produced by SDM-NET enable other applications such as mesh editing, which are not directly supported by the output from other contemporary deep neural networks.

2 RELATED WORK

With the resurgence of deep neural networks, in particular CNNs, and an increasing availability of 3D shape collections [Chang et al. 2015], a steady stream of geometric deep learning methods have been developed for discriminative and generative processing of 3D shapes. In this section, we mainly discuss papers mostly related to

our work, namely deep generative models of 3D shapes, and group them based on the underlying shape representations.

Voxel grids. The direct extension of pixels in 2D images to 3D is the voxel representation, which has a regular structure convenient for CNNs [Girdhar et al. 2016; Qi et al. 2016; Wu et al. 2016]. Variational autoencoders (VAEs) [Kingma and Welling 2013] and Generative Adversarial Networks (GANs) [Goodfellow et al. 2014] can be built with this representation to produce new shapes. Wu et al. [2019] utilize an autoencoder of two branches to encode geometry features and structure features separately, and fuse them into a single latent code to intertwine the two types of features for shape modeling. However, these voxel based representations have huge memory and calculation costs, when the volumetric resolution is high. To address this, sparse voxel-based methods use octrees to adaptively represent the geometry. However, although such adaptive representations can significantly reduce the memory cost, their expressiveness of geometric details is still limited by the resolution of leaf nodes of octrees [Tatarchenko et al. 2017; Wang et al. 2017]. As an improvement, recent work [Wang et al. 2018b] utilizes local planar patches to approximate local geometry in leaf nodes. However, planar patches still have limited capability of describing local geometry, especially for complex local shapes. The patches are in general not smooth or connected, and require further processing, which might degrade the quality of generated shapes.

Multi-view images. To exploit image-like structures while avoiding the high cost of voxels, projecting shapes to multiple 2D views is a feasible approach. Su et al. [2015] project 3D shapes to multi-view images, along with a novel pooling operation for 3D shape recognition. This representation is regular and efficient. However, it does not contain the full 3D shape information. So, although it can be directly used for recognition, additional efforts and processing are needed to reconstruct 3D shapes [Soltani et al. 2017]. It also may not fully recover geometric details due to the incomplete information in multi-view images.

Point clouds. Point clouds have been widely used to represent 3D shapes, since they are flexible and can easily represent the raw data obtained from 3D scanners. The major challenge for deep learning on point clouds is their irregular structure. Qi et al. [2017; 2017] propose PointNet and PointNet++ for 3D classification and segmentation, utilizing pooling operations that are order independent. Yang et al. [2017] exploit an interactive system for segmenting point clouds of indoor scenes. Fan et al. [2017] use point clouds to reconstruct 3D objects from a given image. Achlioptas et al. [2018] introduce a deep autoencoder network for shape representation and generation. However, learning from irregular point clouds is still challenging and their method is only able to produce relatively coarse geometry.

Meshes and multi-chart representations. Deformable modeling of a shape collection, especially of human models [Anguelov et al. 2005; Pons-Moll et al. 2015], operates on meshes with the same connectivity while altering the mesh vertex positions; the shape collection can be viewed as deformations of a template model. For high quality shape generation, especially with large deformations, a manually crafted deformation representation [Gao et al. 2019] is employed

我们网络的架构是一个两级变分自编码器 (VAE) [Kingma和Welling 2013], 它生成结构化可变形网格 (SDM)。在部分级别, 一个PartVAE通过自编码固定连通性、零曲率网格来学习形状部分的变形模型。在结构级别, 我们训练了一个结构化部分VAE (SP-VAE), 它联合学习形状集合的部分结构和部分几何形状, 确保全局形状结构与表面细节之间的协调性。

我们将我们的网络称为SDM-NET, 因为它被训练用于生成结构化可变形网格, 即封闭的可变形网格部件的空间排列, 该排列尊重形状集合 (例如, 形状部件之间的对称性和支撑关系) 的全局部件结构, 例如椅子、飞机等。然而, 我们的网络可以生成具有不同数量部件的形状, 最多可达最大数量。除了上述网格带来的优势外, 结构化表示允许SDM-NET生成的形状立即重用, 例如用于基于装配的建模 [Mitra等人 2013]。此外, 网格部件的可变形性进一步促进了生成形状的编辑和插值。

SDM-NET使用具有一致部件结构的形状集合进行训练, 例如语义分割。然而, 集合中的形状可以是具有任意拓扑和网格连接性的。此类数据集现在已广泛可用, 例如ShapeNet [Chang等人 2015] 和PartNet [Mo等人2019b], 等。虽然SDM-NET的直接输出不是水密网格, 但每个部件都是。

总而言之, 我们工作的主要贡献是:

- 首个能够生成结构化可变形网格的深度生成神经网络。

- 一种对应于双层变分自编码器的创新网络架构, 该架构联合学习形状结构和几何信息。这与近期工作GRASS [Li等人 2017]形成对比, 后者使用独立网络分别学习形状结构和部件几何信息。
- 基于支撑的部件连接优化, 以确保生成合理且物理上有效的形状。

图1展示了我们的SDM-NET在重建具有灵活结构和精细几何细节的形状方面的能力。通过在潜在空间中进行插值, 生成了具有显著结构变化的合理形状。

通过大量实验并与最先进的形状深度生成模型进行比较, 我们证明了SDM-NET在生成高质量网格和形状插值方面的优越性。我们还展示了SDM-NET生成的结构化变形网格能够支持其他应用 (如网格编辑), 而这些应用并非其他当代深度神经网络输出所直接支持。

2 相关工作

随着深度神经网络的复兴, 特别是卷积神经网络 (CNN) 的复兴, 以及三维形状集合 [Chang 等人2015], 的日益增多, 已经开发出大量几何深度学习方法, 用于三维形状的判别和生成处理。在本节中, 我们主要讨论与我们的工作大多相关的论文, 即三维形状的深度生成模型, 并根据其底层的形状表示将它们进行分类。

我们的工作, 即三维形状的深度生成模型, 并根据其底层的形状表示将它们进行分类。

体素网格。将二维图像中的像素直接扩展到三维就是体素表示, 这种结构对CNNs [Girdhar等人2016年; Qi等人2016年; Wu等人 2016]非常方便。变分自编码器 (VAEs) [Kingma和Welling 2013] 以及生成对抗网络 (GANs) [Goodfellow等人 2014] 可以基于这种表示来生成新的形状。Wu等人 [2019] 利用一个具有两个分支的自编码器分别编码几何特征和结构特征, 并将它们融合成一个潜在码, 以将这两种类型的特征交织起来进行形状建模。然而, 当体素分辨率较高时, 这些基于体素的表示具有巨大的内存和计算成本。为了解决这个问题, 稀疏体素方法使用八叉树来自适应地表示几何结构。然而, 尽管这种自适应表示可以显著降低内存成本, 但其几何细节的表达能力仍然受限于八叉树叶节点的分辨率 [Tatarchenko等人2017年; Wang等人 2017]。作为一种改进, 最近的工作 [Wang等人2018b] 利用局部平面块来近似叶节点中的局部几何结构。然而, 平面块在描述局部几何方面仍然能力有限, 特别是对于复杂的局部形状。这些块通常不光滑或不连续, 需要进一步处理, 这可能会降低生成形状的质量。

多视图图像。为了利用图像结构同时避免体素的高成本, 将形状投影到多个二维视图是一种可行的方法。Su等人 [2015] 将3D形状投影到多视图图像中, 并提出了用于3D形状识别的新型池化操作。这

展示了重新, 简洁高效。然而, 它不包含完整的3D形状信息。因此, 尽管它可以直接用于识别, 但仍需要额外的努力 and 处理来重建3D形状 [Soltani等人 2017]。它还可能由于多视图图像中的信息不完整而无法完全恢复几何细节。

点云。点云已被广泛用于表示三维形状, 因为它们灵活且可以轻松表示从三维扫描仪获得的原始数据。深度学习在点云上的主要挑战是它们的非规则结构。Qi 等人 [2017; 2017] 提出了 PointNet 和 PointNet++ 用于三维分类和分割, 利用了顺序无关的池化操作。Yang 等人 [2017] 利用一个交互式系统对室内场景的点云进行分割。Fan 等人 [2017] 使用点云从给定图像中重建三维物体。Achlioptas 等人 [2018] 引入了一个用于形状表示和生成的深度自动编码器网络。然而, 从非规则点云中学习仍然具有挑战性, 并且他们的方法只能产生相对粗糙的几何形状。

网格和多图表示。形状集合的可变形建模, 特别是人类模型 [Anguelov 等人 2005; Pons-Moll 等人 2015], 在操作具有相同连接性的网格时, 会改变网格顶点位置; 形状集合可以看作是模板模型的变形。对于高质量形状生成, 特别是具有较大变形时, 会采用手动设计的变形表示 [Gao 等人 2019]

by [Gao et al. 2018; Tan et al. 2018]. Although these methods can represent and generate shapes with fine details, they require meshes to have the same connectivity. Wang et al. [2018c] reconstruct a mesh-based 3D shape from an RGB image by deforming a sphere-like genus-zero mesh model. Dominic et al. [2018] use a CNN to infer the parameters of free-form deformation (FFD) to deform a template mesh, guided by a target RGB image. Both methods require an image as input to provide guidance, and thus cannot be used for general shape generation tasks without guidance. Moreover, deforming a single mesh limits the topological and geometric complexity of generated shapes.

Multi-chart representations attempt to overcome the above restriction by generating multiple patches that cover a 3D shape. Zhou et al. [2004] create texture atlases with less stretches for texture mapping. Hamu et al. [2018] generate a 3D shape as a collection of conformal toric charts [Maron et al. 2017], each of which provides a cover of the shape with low distortion. Since toric covers are restricted to genus-zero shapes, their multi-chart method still has the same limitation. AtlasNet [Groueix et al. 2018] generates a shape as a collection of patches, each of which is parameterized to a 2D domain as an atlas. While the patches together cover the shape well, visible seams can often be observed. In general, neither the atlases nor the toric charts correspond to meaningful shape parts; the collection is optimized to approximate a shape, but is otherwise unstructured. In contrast, SDM-NET produces structured deformable meshes.

Implicit representations. Several very recent works [Chen and Zhang 2019; Mescheder et al. 2019; Park et al. 2019] show great promise of generative shape modeling using implicit representations. These deep networks learn an implicit function which defines the inside/outside statuses of points with respect to a shape or a signed distance function. The generative models can be applied to various applications including shape autoencoding, generation, interpolation, completion, and single-view reconstruction, demonstrating superior visual quality over methods based on voxels, point clouds, as well as patch-based representations. However, none of these works generate structured or deformable shapes.

Shape structures. Man-made shapes are highly structured, which motivates structure-aware shape processing [Mitra et al. 2013]. Works on learning generative models of 3D shape structures can be roughly divided into two categories [Chaudhuri et al. 2019]: probabilistic graphical models and deep neural networks.

Huang et al. [2015] propose a probabilistic model which computes part templates, shape correspondences, and segmentations from clustered shape collections, and their points in each part are influenced by their correspondence in the template. Similar to [Huang et al. 2015], ShapeVAE [Nash and Williams 2017] generates point coordinates and normals based on different parts, but uses a deep neural network instead of a probabilistic model. Compared to the above two works, our method does not require point-wise correspondences, which can be difficult or expensive to obtain reliably. Moreover, our method encodes both global spatial structure like support relations, and local geometric deformation, producing shapes with reasonable structures and fine details.

Li et al. [2017] introduce GRASS, a generative recursive autoencoder for shape structures, based on Recursive Neural Networks (RvNNs). Like SDM-NET, GRASS also decouples structure and geometry representations. However, a key difference is that SDM-NET *jointly* encodes global shape structure and part geometry, while GRASS trains *independent* networks for structure and part geometry generations. In terms of outputs, GRASS generates a *hierarchical* organization of bounding boxes, and then fills them with voxel parts. SDM-NET produces a *set* of shape parts, each of which is a deformable mesh to better capture finer surface details. Lastly, the structural autoencoder of GRASS requires symmetry hierarchies for training while SDM-NET only requires consistent semantic segmentation and employs the support information to produce shapes with support stability.

Concurrent to SDM-NET, Mo et al. [2019a] develop *StructureNet*, which learns a generative autoencoder of shape structures based on graph neural networks. StructureNet shares much commonality with GRASS but extends it in two important ways. First, unlike GRASS, which is limited to encoding binary trees, StructureNet can directly encode shapes represented as n -ary graphs, aimed to facilitate a consistent hierarchical representation of shapes within the same category. Second, StructureNet also accounts for horizontal inter-part relationships between siblings. The outputs from StructureNet are either box structures or point cloud shapes. Our SDM-Net analyzes and encodes shape structures by not only using the consistent representation across the same shape families but also with support stability. In addition, it is expected our mesh-based representation with deformable parts is able to capture more geometry details than box and point cloud based representations adopted by StructureNet.

3 METHODOLOGY

Overview. Given a collection of shapes of the same category with part-level labels, our method represents them using a structured set of deformable boxes, each corresponding to a part. The pioneering works [Kim et al. 2013; Ovsjanikov et al. 2011] have shown the representation power of using a collection of boxes to analyze and explore shape collections. However, it is highly non-trivial to extend their techniques to shape generation, since boxes are generally of a coarse representation. We tackle this challenge by allowing individual boxes to be flexibly deformable and propose a two-level VAE architecture called SDM-NET, including PartVAE for encoding the geometry of deformable boxes, and SP-VAE for joint encoding of part geometry and global structure such as symmetry and support. Moreover, to ensure that decoded shapes are physically plausible and stable, we introduce an optimization based on multiple constraints including support stability, which can be compactly formulated and efficiently optimized. Our SDM-NET model allows easy generation of plausible meshes with flexible structures and fine details.

We first introduce the encoding of each part, including both the geometry and structure information. We then introduce our SDM-NET involving VAEs at both the local part geometry level (PartVAE), and global joint embedding of structure and geometry (SP-VAE). Then we briefly describe how the support relationships are extracted,

由 [Gao 等人 2018; Tan 等人 2018]采用。尽管这些方法可以表示和生成具有精细细节的形状，但它们要求网格具有相同的连接性。Wang 等人 [2018c] 通过变形球状零亏格网格模型，从 RGB 图像重建基于网格的 3D 形状。Dominic 等人 [2018] 使用 CNN 推断自由形式变形 (FFD) 的参数来变形模板网格，并由目标 RGB 图像进行引导。这两种方法都需要图像作为输入来提供指导，因此在没有指导的情况下不能用于通用的形状生成任务。此外，变形单个网格限制了生成形状的拓扑和几何复杂性。

多图表表示方法试图通过生成覆盖3D形状的多个补丁来克服上述限制。Zhou等人 [2004] 创建了用于纹理映射的纹理图集，拉伸较少。Hamu等人 [2018] 将3D形状生成为一组共形环面图表 [。Maron等人 2017],，每个图表都为形状提供低失真的覆盖。由于环面覆盖仅限于零曲率形状，他们的多图表方法仍然存在相同的限制。AtlasNet [Groueix等人 2018] 将形状生成为一组补丁，每个补丁参数化为一个2D域作为图集。虽然这些补丁一起很好地覆盖了形状，但通常可以看到可见的接缝。一般来说，无论是图集还是环面图表，都与有意义的形状部分不对应；该集合被优化以近似形状，但除此之外是无结构的。相比之下，SDM-NET生成结构化可变形网格。

隐式表示。最近几项研究 [Chen 和 Zhang 2019; Mescheder 等人 2019; Park 等人 2019]表明，使用隐式表示进行生成式形状建模具有巨大潜力。这些深度网络学习隐式函数，该函数定义了点相对于形状或有符号距离函数的内部/外部状态。生成模型可应用于各种应用，包括形状自动编码、生成、插值、补全和单视图重建，在基于体素、点云以及基于补丁的表示方法上展现出更优越的视觉效果。然而，这些作品中没有生成结构化或可变形形状的。

形状结构。人工形状具有高度结构化，这推动了结构感知形状处理 [Mitra 等人 2013]。关于学习 3D 形状结构生成模型的研究大致可分为两类 [Chaudhuri 等人 2019]: 概率图模型和深度神经网络。

黄等人. [2015]提出了一种概率模型，该模型从聚类形状集合中计算部分模板、形状对应关系和分割，并且每个部分中的点受其模板中对应关系的影响。类似于 [黄等人. 2015], ShapeVAE [Nash和Williams 2017] 基于不同的部分生成点坐标和法线，但使用深度神经网络而不是概率模型。与上述两项工作相比，我们的方法不需要点对应关系，而点对应关系可能难以可靠地获取或成本高昂。此外，我们的方法编码了全局空间结构（如支持关系）和局部几何变形，生成了具有合理结构和精细细节的形状。

李等人. [2017] 介绍了GRASS，一种基于递归神经网络（RvNNs）的形状结构的生成式递归自动编码器。类似于 SDM-NET，GRASS也解耦了结构和几何表示。然而，一个关键区别是SDM-NET联合编码全局形状结构和部分几何，而 GRASS为结构和部分几何生成训练独立的网络。在输出方面，GRASS生成边界框的层次组织，然后用体素部分填充它们。SDM-NET生成一组形状部分，每个部分都是一个可变形网格，以更好地捕捉更精细的表面细节。最后，GRASS的结构自动编码器需要对称层次进行训练，而SDM-NET只需要一致的语义分割，并利用支持信息来生成具有支持稳定性的形状。

与SDM-NET并行，Mo等人 [2019a]开发了StructureNet，该网络基于图神经网络学习形状结构的生成自编码器。StructureNet与GRASS具有许多共性，但在两个方面进行了扩展。首先，与仅限于编码二叉树的GRASS不同，StructureNet可以直接编码表示为n元图的形状，旨在促进同一类别形状之间的一致分层表示。其次，StructureNet还考虑了兄弟节点之间的水平部分关系。StructureNet的输出要么是框结构，要么是点云形状。我们的SDM-Net通过不仅使用相同形状家族之间的一致表示，而且还支持稳定性来分析和编码形状结构。此外，我们期望基于网格的、具有可变形部分的表示能够比StructureNet采用的基于框和点云的表示捕获更多几何细节。

3 方法论

概述。给定一组具有部分级标签的同类形状集合，我们的方法使用一组结构化的可变形框来表示它们，每个框对应一个部分。开创性的工作 [Kim 等人 2013; Ovsjanikov 等人 2011] 已经展示了使用框集合来分析和探索形状集合的表示能力。然而，将他们的技术扩展到形状生成是非常困难的，因为框通常表示粗糙。我们通过允许单个框灵活地变形来应对这一挑战，并提出了一种名为 SDM-NET 的双层 VAE 架构，包括 PartVAE 用于编码可变形框的几何形状，以及 SP-VAE 用于联合编码部分几何形状和全局结构（例如对称性和支撑）。此外，为了确保解码的形状是物理上合理且稳定的，我们引入了一种基于多个约束（包括支撑稳定性）的优化方法，该方法可以紧凑地表述并高效地优化。我们的 SDM-NET 模型允许轻松生成具有灵活结构和精细细节的合理网格。

我们首先介绍每个部分的编码，包括几何和结构信息。然后我们介绍我们的SDM-NET，它涉及在局部部分几何级别（PartVAE）和结构及几何的全局关节嵌入（SP-VAE）处的VAE。接着我们简要描述如何提取支持关系，

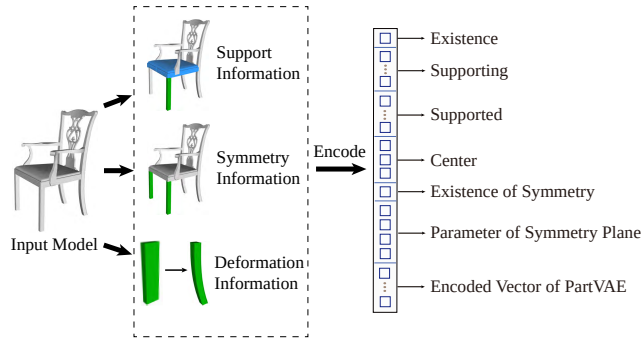


Fig. 2. Encoding of each shape part, including its structure and geometry information. The former includes both the support and symmetry information, and the latter describes the deformation of the bounding box, compactly represented as the latent vector of PartVAE. Detailed explanation of each entry of the code is presented in Section 3.1.

and finally present our optimization for generating plausible and well supported shapes.

3.1 Encoding of a Shape Part

Based on semantic part-level labels, a shape is decomposed into a set of parts. Each part is represented using a deformable bounding box, as illustrated in Figure 3. Let n be the total number of part labels that appear across different shapes in the specified object category. For a given shape, it may contain a fewer number of parts as some parts may not be present. To make analysis and relationship representation easier, we assume the initial bounding box (before deformation) of each part is axis aligned. This is sufficient in practice, since each bounding box is allowed to have substantial deformation to fit the target geometry. The initial bounding primitive being a box does not prevent the internal part geometry from being complex, since geometric details can be captured and preserved through non-rigid registration (see Section 3.2 for details). Without loss of generality, the bounding boxes are used in our framework.

The geometry and associated relationships of each part are encoded by a representation vector $\mathbf{rv}$, as illustrated in Figure 2. The detailed definition of this vector is given as follows.

- $\mathbf{rv}_1 \in \{0, 1\}$ indicates the existence of this part.
- $\mathbf{rv}_2 \in \{0, 1\}^n$ is a vector with n dimensions to indicate which parts are supported by this part.
- $\mathbf{rv}_3 \in \{0, 1\}^n$ is a vector with n dimensions to indicate which parts support the current part.
- $\mathbf{rv}_4 \in \mathbb{R}^3$ is the 3D position of the bounding box center.
- $\mathbf{rv}_5 \in \{0, 1\}$ indicates the existence of a symmetric part.
- $\mathbf{rv}_6 \in \mathbb{R}^4$ records the parameters $a-d$ of the symmetry plane represented in an implicit form, i.e., $ax + by + cz + d = 0$.
- $\mathbf{rv}_7$ is the encoded vector from the PartVAE described in Section 3.2, which encodes its geometry. By default, $\mathbf{rv}_7 \in \mathbb{R}^{64}$.

The ID of each part, used in $\mathbf{rv}_2$ and $\mathbf{rv}_3$, is pre-determined and stored in advance for the dataset. Each value in $\mathbf{rv}_1$, $\mathbf{rv}_2$, $\mathbf{rv}_3$ and $\mathbf{rv}_5$ is 1 if exists and 0 otherwise. For generated vectors, we treat a value above 0.5 as true and below as false. The length of this

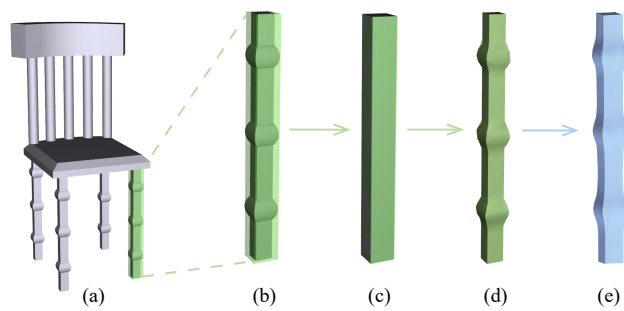


Fig. 3. An example of representing a chair leg with detailed geometry by a deformable bounding box. (a) a chair with one of its leg parts highlighted, (b) the targeted part in (a) with the bounding box overlaid, (c) the bounding box used as the template, (d) deformed bounding box after non-rigid registration, (e) recovered shape using PartVAE.

vector is $2n + 73$ and between 77 and 101 for all the examples in this paper. Note that other information such as the label of the part that is symmetric to the current one (if exists) is fixed for a collection (e.g. the right armrest of a chair is symmetric to the left armrest of the chair) and therefore not encoded in the vector. In our current implementation, we only consider reflection symmetry and keep one symmetric component (if any) for each part. Although this is somewhat restrictive, it is very common and sufficient to cope with most cases. In practice, we first perform global reflection symmetry detection [Podolak et al. 2006] to identify components that are symmetric to each other w.r.t. a symmetry plane. This is then supplemented by local reflection symmetry detection by checking if pairs of parts have reflective symmetry.

3.2 PartVAE for Encoding Part Geometry

For each part, the axis-aligned bounding box (AABB) is first calculated. The bounding box of the same part type provides a uniform domain across different shapes, and the geometry variations are viewed as different deformation functions applied to the same domain. We take a common template, namely a unit cube mesh box_0 with 19.2K triangles, to represent each part. We first translate and scale it to fit the bounding box of the part. Denote by $b_{i,j}$ the bounding box transformed from box_0 for the j^{th} part $c_{i,j}$ on the i^{th} shape s_i . We treat it as initialization, and apply non-rigid coarse-to-fine registration [Zollhöfer et al. 2014], which deforms $b_{i,j}$ to $b'_{i,j}$, as illustrated in Figure 3 (d). $b'_{i,j}$ shares the geometry details with the part $c_{i,j}$ and has the same mesh connectivity as the unit cube box box_0 .

The variational autoencoder has been used to encode the geometric priors for point cloud segmentation [Meng et al. 2019] and mesh generation [Tan et al. 2018]. Similar to [Tan et al. 2018], using meshes of the same connectivity makes it feasible to build a variational autoencoder to represent the deformation of $b'_{i,j}$. The convolutional VAE architecture in [Gao et al. 2018] is employed for compactly representing plausible deformation of each part, allowing new variations to be synthesized. The architecture is shown in Figure 4. The input is a $V \times 9$ dimensional matrix, where V is the

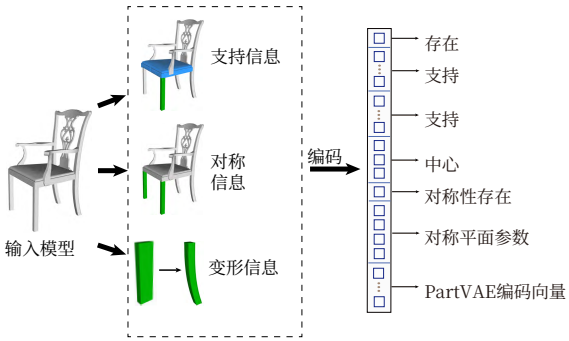


图2. 每个形状部分的编码，包括其结构和几何信息。前者包括支撑和对称信息，后者描述了边界框的变形，紧凑地表示为PartVAE的潜在向量。代码的每个条目的详细解释见第3.1节。

最后展示我们的优化方法，用于生成合理且得到充分支持的结构。

3.1 形状部分的编码

基于语义部分级标签，形状被分解为一组部分。每个部分使用可变形边界框表示，如图3所示。令 n 为指定对象类别中不同形状出现的部分标签总数。对于一个给定的形状，它可能包含较少的部分，因为某些部分可能不存在。为了便于分析和关系表示，我们假设每个部分的初始边界框（变形前）是轴对齐的。这在实践中是足够的，因为每个边界框允许有显著的变形以适应目标几何形状。初始边界基元是框，这并不妨碍内部部分几何形状的复杂性，因为几何细节可以通过非刚性配准来捕获和保留（有关详细信息，请参见第3.2节）。不失一般性，我们的框架中使用边界框。

每个部件的几何形状及其相关关系由表示向量 $\mathbf{rv}$ 编码，如图2所示。该向量的详细定义如下。

- $\mathbf{rv}_1 \in \{0, 1\}$ 表示该部件的存在。
- $\mathbf{rv}_2 \in \{0, 1\}^n$ 是一个 n 维向量，用于指示哪些部件由该部件支持。
- $\mathbf{rv}_3 \in \{0, 1\}^n$ 是一个 n 维向量，用于指示哪些部件支持当前部件。

- $\mathbf{rv}_4 \in \mathbb{R}^3$ 是边界框中心的3D位置。
- $\mathbf{rv}_5 \in \{0, 1\}$ 表示对称部件的存在。
- $\mathbf{rv}_6 \in \mathbb{R}^4$ 记录了对称平面的参数 $a-d$ ，该平面以隐式形式表示，即 $ax + by + cz + d = 0$ 。
- $\mathbf{rv}_7$ 是从第 3.2 节中描述的PartVAE编码的向量。它编码了其几何形状。默认情况下， $\mathbf{rv}_7 \in \mathbb{R}^{64}$ 。

每个部分的 ID，用于 $\mathbf{rv}_2$ 和 $\mathbf{rv}_3$ ，是预先确定并提前存储在数据集中的。 $\mathbf{rv}_1$ 、 $\mathbf{rv}_2$ 、 $\mathbf{rv}_3$ 和 $\mathbf{rv}_5$ 中的每个值如果存在则为 1，否则为 0。对于生成的向量，我们将大于 0.5 的值视为真，小于则视为假。这个长度

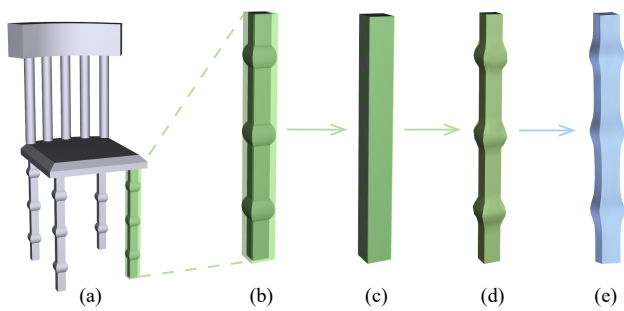


图3. 一个使用可变形边界框表示椅子腿的详细几何形状的示例。(a) 一把椅子，其中一个腿部分被突出显示，(b)在(a)中目标部分上叠加边界框，(c)用作模板的边界框，(d)非刚性配准后的变形边界框，(e)使用PartVAE恢复的形状。

虽然这有点限制，但对于大多数情况非常常见且足够。在实践中，我们首先执行全局反射对称检测 [Podolak等人 2006] 来识别相对于对称平面相互对称的部件。然后通过检查部件对是否具有反射对称性来补充局部反射对称检测。

3.2 PartVAE用于编码部件几何形状

对于每个部件，首先计算轴对齐边界框（AABB）。相同部件类型的边界框在不同形状之间提供了一个统一的域，几何变化被视为应用于同一域的不同变形函数。我们采用一个通用模板，即具有19.2K三角形的单位立方体网格框 box_0 ，来表示每个部件。我们首先将其平移和缩放以适应部件的边界框。将第 i 个形状 s_i 上的第 j 个部件 $c_{i,j}$ 从框 box_0 转换后的边界框记为 $b_{i,j}$ 。我们将其视为初始化，并应用非刚性粗到细配准 [Zollhöfer等人 2014]，将 $b_{i,j}$ 变形为 $b'_{i,j}$ ，如图3 (d) 所示。 $b'_{i,j}$ 与部件 $c_{i,j}$ 共享几何细节，并且与单位立方体框 box_0 具有相同的网格连接性。

变分自编码器已被用于对点云分割进行几何先验编码 [孟等人 2019] 以及网格生成 [谭等人 2018]。类似于 [谭等人 2018]，使用相同连接性的网格，使得构建一个变分自编码器来表示 $b'_{i,j}$ 的变形成为可能。在 [高等人 2018] 中采用的卷积VAE架构用于紧凑地表示每个部分的合理变形，允许合成新的变化。该架构如图4所示。输入是一个 $V \times 9$ 维矩阵，其中 V 是模板边界框网格的顶点数。矩阵的每一行是一个9维向量，它表征了每个顶点1环邻域的局部变形，包括旋转轴、旋转角和缩放因子。它通过两个卷积层和一个全连接层来获得均值和方差。解码器镜像编码器的结构以恢复变形表示，但具有不同的可训练权重。由于每种部分类型都有其自己的特征，我们为跨不同形状具有相同部分类型的所有部分训练一个PartVAE。

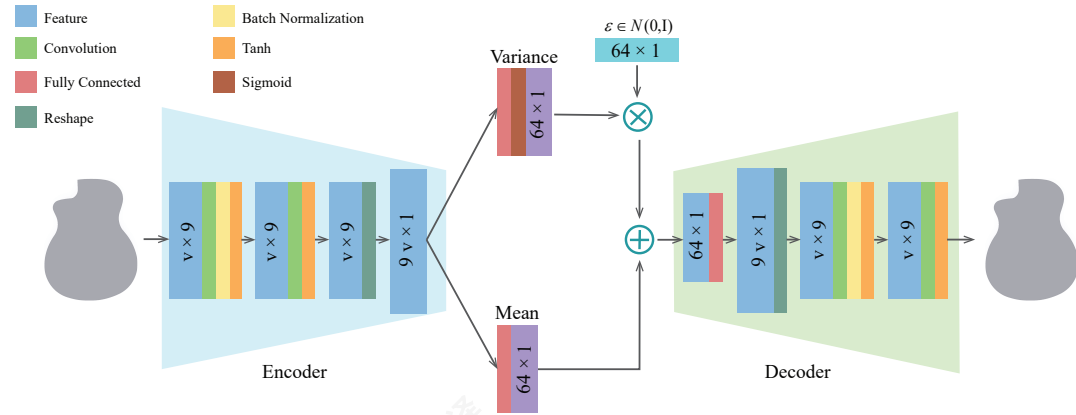


Fig. 4. Architecture of PartVAE for encoding the geometry details of a part represented as the deformation of the associated template box. V is the number of vertices in the template box. $N(0, I)$ is the Gaussian distribution with 0 mean and identity covariance.

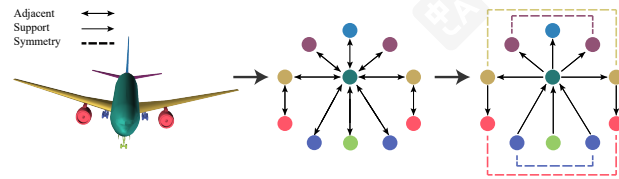


Fig. 5. Illustration of support and symmetry relations between parts of an airplane model. The support relations, detected by the approach in [Huang et al. 2016], turn an undirected adjacency graph (Middle) to a directed support graph (Right).

number of vertices for the template bounding box mesh. Each row of the matrix is a 9-dimensional vector that characterizes the local deformation of 1-ring neighborhood of each vertex including the rotation axis, rotation angle and scaling factor. It passes through two convolutional layers followed by a fully connected layer to obtain the mean and variance. The decoder mirrors the structure of the encoder to recover the deformation representation, but with different trainable weights. Since each part type has its own characteristics, we train a PartVAE for all the parts with the same part type across different shapes.

3.3 Supporting Structure Analysis

Structure captures the relationships between parts, and proper encoding of shape structure is crucial to generate plausible shapes. Symmetry as one of the structural relationships has been well explored, for example effectively used in GRASS [Li et al. 2017]. Besides symmetry, support relationships have been demonstrated useful for structural analysis to synthesize physically plausible new structures with support and stability [Huang et al. 2016]. Compared to symmetry, support relationships provide a more direct mechanism to model the relations between adjacent parts. We thus use both symmetry and support to encode the structural relationships between parts (Section 3.1). Note that our work is the first attempt to encode support-based shape structures in deep neural networks.

Following [Huang et al. 2016], we detect the support relations between adjacent parts as one of three support sub-structures, namely, “support from below”, “support from above”, and “support from side”. As illustrated in Figure 5, the detected support relations turn an undirected adjacency graph to a directed support graph. For each detected support relation of a part, we encode the labels of its supported and supporting parts in our part feature vector (Section 3.1). Our feature coding is flexible to represent the cases including one part being supported by multiple parts, as well as multiple parts being supported by one part. Since for all the cases in our shape dataset, the sub-structure type for each support relation between two adjacent parts is fixed, the support sub-structure types are kept in a look-up table but not encoded in our part feature vector. Given the supporting and supported part labels from a decoded part feature vector, we can efficiently obtain the corresponding sub-structures from this look-up table.

3.4 SP-VAE for Structured Deformable Mesh Encoding

We build SP-VAE to jointly encode the structure of a shape represented as the layout of boxes, and the geometry of its parts. By analyzing their joint distribution, it helps ensure that the geometry of the generated shape is coherent with the structure and the geometries of individual parts are consistent (i.e., of compatible styles). Our SP-VAE takes the concatenation of representation vectors for all the parts as input (see Section 3.1). It encodes parts in a consistent order during encoding and decoding. This concatenated vector covers both the geometric details of individual parts encoded using PartVAE, and the relationships between them. The SP-VAE uses multiple fully connected layers, and the architecture is illustrated in Figure 6.

Let $Enc_S(\cdot)$ and $Dec_S(\cdot)$ denote the encoder and decoder of our SP-VAE network, respectively. x represents the input concatenated feature vector of a shape, $\tilde{x} = Enc_S(x)$ is the encoded latent vector, and $x' = Dec_S(\tilde{x})$ is the reconstructed feature vector. Our SP-VAE minimizes the following loss:

$$L_{SP-VAE} = \lambda_1 L_{recon} + \lambda_2 L_{KL} + L_{RegVAE}, \quad (1)$$

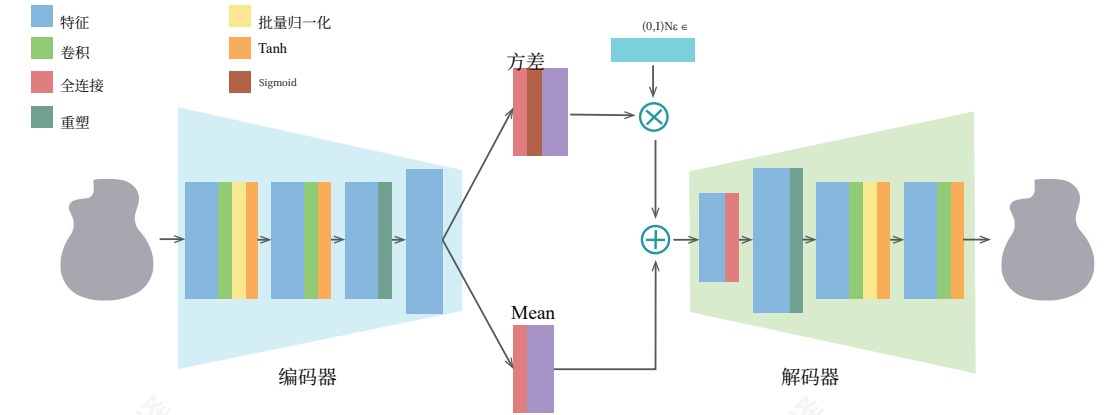


图4. PartVAE的架构，用于编码表示为相关模板框变形的部分的几何细节。V是模板框中的顶点数。N(0, I)是均值为0、协方差为单位矩阵的高斯分布。

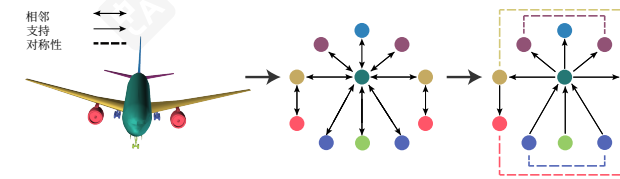


图5. 飞机模型各部分之间的支持和对称关系示意图。支持关系，通过 [Huang等人 2016] 的方法检测，将无向邻接图（中间）转换为有向支持图（右侧）。

该矩阵的每一行是一个9维向量，它表征了每个顶点1环邻域的局部变形，包括旋转轴、旋转角和缩放因子。它通过两个卷积层和一个全连接层来获得均值和方差。解码器镜像编码器的结构以恢复变形表示，但具有不同的可训练权重。由于每种部分类型都有其自己的特征，我们为跨不同形状具有相同部分类型的所有部分训练一个PartVAE。

3.3 支持结构分析

结构捕捉了部分之间的关系，而形状结构的正确编码对于生成合理的形状至关重要。对称性作为结构关系之一已被广泛研究，例如在GRASS [Li等人 2017]的工作中有效应用。除了对称性，支撑关系已被证明对结构分析有用，可以合成具有支撑和稳定性的物理上合理的结构 [Huang等人 2016]。与对称性相比，支撑关系提供了更直接的方式来建模相邻部分之间的关系。因此，我们使用对称性和支撑关系来编码部分之间的结构关系（第3.1节）。请注意，我们的工作首次尝试在深度神经网络中编码基于支撑的形状结构。

遵循 [Huang等人 2016]，我们将相邻部分之间的支撑关系检测为三种支撑子结构之一，即“下方支撑”、“上方支撑”和“侧面支撑”。如图5所示，检测到的支撑关系将无向邻接图转换为有向支撑图。对于每个检测到的部分支撑关系，我们在其部分特征向量中编码其被支撑部分和支持部分的标签（第3.1节）。我们的特征编码是灵活的，可以表示包括一个部分被多个部分支撑，以及多个部分被一个部分支撑的情况。由于在我们的形状数据集中，两个相邻部分之间的每个支撑关系的子结构类型是固定的，因此支撑子结构类型保存在查找表中，而不是编码在我们的部分特征向量中。根据解码的部分特征向量中的支撑部分和被支撑部分的标签，我们可以从该查找表中高效地获取相应的子结构。

3.4 SP-VAE用于结构化可变形网格编码

我们构建SP-VAE以联合编码表示为盒子布局的形状结构，及其部件的几何形状。通过分析它们的联合分布，有助于确保生成的形状的几何形状与结构一致，并且各个部件的几何形状（即风格兼容）保持一致。我们的SP-VAE将所有部件的表示向量的连接作为输入（见第3.1节）。它在编码和解码过程中以一致的顺序编码部件。这个连接向量既包含了使用PartVAE编码的各个部件的几何细节，也包含了它们之间的关系。SP-VAE使用多个全连接层，其架构如图6所示。

$Enc_S(\cdot)$ 和 $Dec_S(\cdot)$ 分别表示我们的 SP-VAE 网络的编码器和解码器。x 表示形状的输入拼接特征向量， $\tilde{x} = Enc_S(x)$ 是编码的潜在向量， $x' = Dec_S(\tilde{x})$ 是重建的特征向量。我们的 SP-VAE 最小化以下损失：

$$L_{SP-VAE} = \lambda_1 L_{recon} + \lambda_2 L_{KL} + L_{RegVAE}, \quad (1)$$

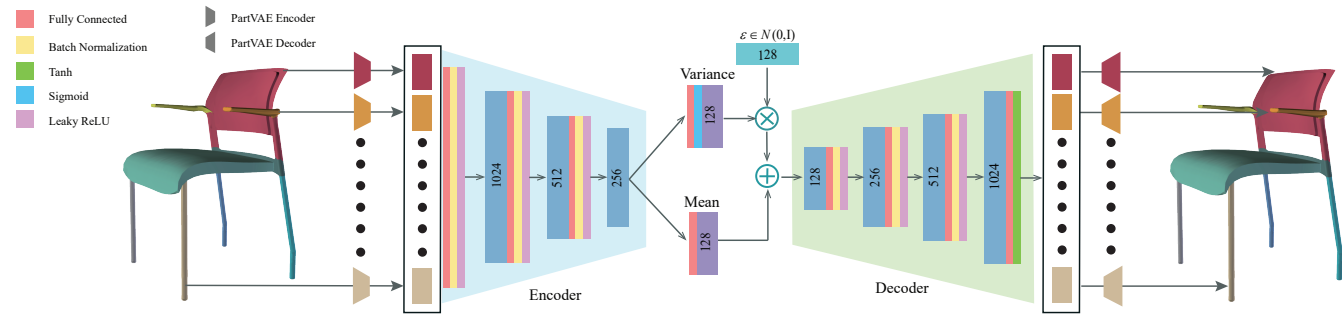


Fig. 6. Architecture of Structured Parts VAE (SP-VAE). The detailed geometry is encoded by PartVAE. The support- and symmetry-induced structure and the associated latent code of PartVAE are encoded by the Structured Parts VAE.

where λ_1 and λ_2 are the weights of different loss terms, and

$$L_{recon} = \frac{1}{N} \sum_{\mathbf{x} \in \mathcal{S}} \|\mathbf{x} - \mathbf{x}'\|_2^2 \quad (2)$$

denotes the MSE (mean squared error) reconstruction loss to ensure better reconstruction. Here $\mathcal{S}$ is the training dataset and $N = |\mathcal{S}|$ is the number of shapes in the training set.

$$L_{KL} = D_{KL}(\hat{q}(\tilde{\mathbf{x}}|\mathbf{x})|\hat{p}(\tilde{\mathbf{x}})) \quad (3)$$

is the KL divergence to promote Gaussian distribution in the latent space, where $\hat{q}(\tilde{\mathbf{x}}|\mathbf{x})$ is the posterior distribution given feature vector $\mathbf{x}$, and $\hat{p}(\tilde{\mathbf{x}})$ is the Gaussian prior distribution. L_{RegVAE} is the squared ℓ_2 norm regularization term of the network parameters used to avoid overfitting. The Gaussian distribution makes it effective to generate new shapes by sampling in the latent space, which is used for random generation and interpolation.

3.5 Shape Generation and Refinement

The latent space of SP-VAE provides a meaningful space for shape generation and interpolation. Extensive experimental results are shown in Section 5. Random sampling in the latent space can generate novel shapes. However, although the desired geometry and structure from the decoded feature vector are generally reasonable, they may not satisfy supporting and physical constraints exactly, resulting in shapes which may include parts not exactly in contact, or may be unstable. Inspired by [Averkiou et al. 2014], we propose to use an effective global optimization to refine the spatial relations between parts by mainly using the associated symmetry and support information.

Denote the center position and size (half of the length in each dimension) of the i^{th} part as $\mathbf{p}_i$ and $\mathbf{q}_i$, each being a 3-dimensional vector corresponding to x , y and z axes, where $\mathbf{p}_i$ is directly obtained from the representation vector, and $\mathbf{q}_i$ is determined by the bounding box after recovering the part geometry. Denote by $\mathbf{p}'_i$ and $\mathbf{q}'_i$ the position and size of the i^{th} part after global optimization. The objective of this optimization is to minimize the changes between the optimized position/scale and the original position/scale

$$\sum_i \|\mathbf{p}'_i - \mathbf{p}_i\|^2 + \alpha \|\mathbf{q}'_i - \mathbf{q}_i\|^2, \quad (4)$$

while ensuring the following constraints are satisfied. α is a weight to balance the two terms, and is fixed to 10 in our experiments. The symmetry and equal length constraints are from [Averkiou et al. 2014], though we use the support relationships to help identify equal length constraints more reliably. The remaining constraints are unique in our approach. Figure 7 illustrates typical problematic cases which are handled by our refinement optimization.

Symmetry Constraint: If the generated i^{th} part has the symmetry indicator flagged in its representation vector, its symmetry part (denoted using index j) also exists. Let $\mathbf{n}_i$ and d_i denote the normal and the intercept of the symmetry plane, respectively. Enforcing the symmetry constraint leads to the following constraints to be satisfied: $\frac{(\mathbf{p}'_i - \mathbf{p}'_j)}{2} \cdot \mathbf{n}_i + d_i = 0$, $(\mathbf{p}'_i - \mathbf{p}'_j) \times \mathbf{n}_i = 0$. The symmetry of two parts is viewed as an undirectional relationship. If the symmetry indicator (rv5 of the representation vector) of either part i or j is 1, we consider these two parts as symmetric.

Equal Length Constraint: A set of parts which are simultaneously supported by a common part, and simultaneously support another common part, are considered as a group to have the same length along the supporting direction. For this purpose, the ground is considered as a virtual part. For example, the four legs of a table supporting the same table top part and at the same time being supported by the ground should have the same height. These can be easily detected by traversing the support structure. An example that violates this constraint is illustrated in Figure 7 (a). The equal length constraints can be formulated as $\mathbf{q}'_i[t] = \mathbf{q}'_k[t]$, $k \in g_i$, where g_i is an equal-length group containing the i^{th} part, and t is the supporting direction. $t = 0, 1, 2$ respectively represents x , y and z directions where y is the upright direction.

Support Relationship Constraint: In order for a supporting part to well support a part being supported, two requirements are needed: 1) in the supporting direction, the bounding box of the supporting part should have tangential relation (or a small amount of intersection in practice) with the bounding box of the part being supported (see Figure 7 (e) for a problematic case violating this constraint). If the i^{th} part supports the j^{th} part, the following inequality should be satisfied along the supporting direction. $\mathbf{p}_j[t] - \mathbf{q}'_j[t] + \epsilon \mathbf{q}'_j[t] \leq \mathbf{p}'_i[t] + \mathbf{q}'_i[t] \leq \mathbf{p}_j[t] - \mathbf{q}'_j[t] + 2\epsilon \mathbf{q}'_j[t]$, where t is the supporting direction, and ϵ controls the amount of overlap

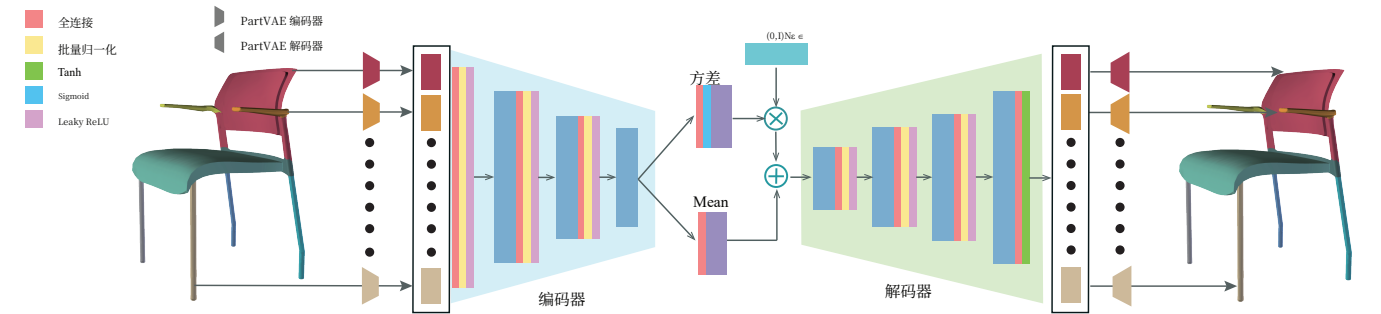


图6. 结构化部件变分自编码器 (SP-VAE) 的架构。部件变分自编码器编码了详细的几何结构。部件变分自编码器由支撑和对称诱导的结构及其相关的潜在码被结构化部件变分自编码器编码。

其中 λ_1 和 λ_2 是不同损失项的权重, 并且

$$L_{recon} = \frac{1}{N} \sum_{\mathbf{x} \in \mathcal{S}} \|\mathbf{x} - \mathbf{x}'\|_2^2 \quad (2)$$

表示均方误差 (MSE) 重建损失, 以确保更好的重建效果。其中 $\mathcal{S}$ 是训练数据集, $N = |\mathcal{S}|$ 是训练集中形状的数量。

$$L_{KL} = D_{KL}(\hat{q}(\tilde{\mathbf{x}}|\mathbf{x})|\hat{p}(\tilde{\mathbf{x}})) \quad (3)$$

是用于在潜在空间中促进高斯分布的KL散度, 其中 $\hat{q}(\tilde{\mathbf{x}}|\mathbf{x})$ 是给定特征向量 $\mathbf{x}$ 的后验分布, $\hat{p}(\tilde{\mathbf{x}})$ 是高斯先验分布。LRegVAE 是网络参数的平方L2范数正则化项, 用于避免过拟合。高斯分布使得在潜在空间中采样生成新形状非常有效, 这可用于随机生成和插值。

3.5 形状生成与优化

SP-VAE 的潜在空间为形状生成和插值提供了有意义的空间。第5节展示了大量的实验结果。在潜在空间中随机采样可以生成新颖的形状。然而, 尽管解码特征向量得到的几何形状和结构通常合理, 但它们可能不完全满足支撑和物理约束, 导致形状中可能存在部分未完全接触或可能不稳定的情况。受 [Averkiou et al. 2014] 的启发, 我们提出使用有效的全局优化方法, 主要通过利用相关的对称性和支撑信息来优化部件之间的空间关系。

将第 i 个部件的中心位置和大小 (每个维度长度的一半) 表示为 $\mathbf{p}_i$ 和 $\mathbf{q}_i$, 它们都是对应于 x 、 y 和 z 轴的三维向量, 其中 $\mathbf{p}_i$ 直接从表示向量中获取, 而 $\mathbf{q}_i$ 则通过恢复部件几何形状后的边界框确定。将第 i 个部件在全局优化后的位置和大小表示为 $\mathbf{p}'_i$ 和 $\mathbf{q}'_i$ 。该优化的目标是 minimized 优化位置/尺寸与原始位置/尺寸之间的变化, 同时确保满足以下约束条件。{v1} 是一个用于平衡这两个项的权重, 在我们的实验中固定为 10。对称性和等长约束来自 [v2] Averkiou 等人 [v3], 尽管我们使用支持关系来更可靠地识别等长约束。其余约束是我们方法中的独特之处。图 7 说明了由我们的细化优化处理的典型问题案例。

$$\sum_i \|\mathbf{p}'_i - \mathbf{p}_i\|^2 + \alpha \|\mathbf{q}'_i - \mathbf{q}_i\|^2, \quad (4)$$

同时确保满足以下约束条件。 α 是一个用于平衡这两个项的权重, 在我们的实验中固定为 10。对称性和等长约束来自 [Averkiou 等人 2014], 尽管我们使用支持关系来更可靠地识别等长约束。其余约束是我们方法中的独特之处。图 7 说明了由我们的细化优化处理的典型问题案例。

对称约束: 如果生成的第 i 个部件在其表示向量中标记了对称指示器, 则其对称部分 (使用索引 j 表示) 也存在。令 $\mathbf{n}_i$ 和 d_i 分别表示对称平面的法向量和截距。强制执行对称约束会导致以下约束条件需要满足: $\frac{(\mathbf{p}'_i - \mathbf{p}'_j)}{2} \cdot \mathbf{n}_i + d_i = 0$, $(\mathbf{p}'_i - \mathbf{p}'_j) \times \mathbf{n}_i = 0$, 两个部件的对称性被视为一种无方向的关系。如果部件 i 或 j 的对称指示器 (表示向量的 rv5) 为 1, 我们将认为这两个部件是对称的。

等长约束: 一组同时被共同部件支撑, 并同时支撑另一个共同部件的部件, 被认为在支撑方向上具有相同的长度。为此, 地面被视为一个虚拟部件。例如, 支撑相同桌面部件并同时被地面支撑的桌子的四条腿应该具有相同的高度。这些可以通过遍历支撑结构来轻松检测。图 7 (a) 中所示的是一个违反此约束的示例。等长约束可以表示为 $\mathbf{q}'_i[t] = \mathbf{q}'_k[t]$, $k \in g_i$, 其中 g_i 是一个包含第 i 个部件的等长组, t 是支撑方向。 $t = 0, 1, 2$, 分别表示 x 、 y 和 z 方向, 其中 y 是垂直方向。

支持关系约束: 为了让支撑部件能够良好地支撑被支撑部件, 需要两个要求: 1) 在支撑方向上, 支撑部件的包围盒应与被支撑部件的包围盒有切线关系 (在实际中可以有少量交集) (见图 7 (e) 所示的一个违反此约束的问题案例)。如果第 i 个部件支撑 j^{th} 部件, 则沿支撑方向应满足以下不等式 $\mathbf{p}'_j[t] - \mathbf{q}'_j[t] + \epsilon \mathbf{q}'_j[t] \leq \mathbf{p}'_i[t] + \mathbf{q}'_i[t] \leq \mathbf{p}'_j[t] - \mathbf{q}'_j[t] + 2\epsilon \mathbf{q}'_j[t]$, 其中 t 是支撑方向, 而 ϵ 控制重叠的量

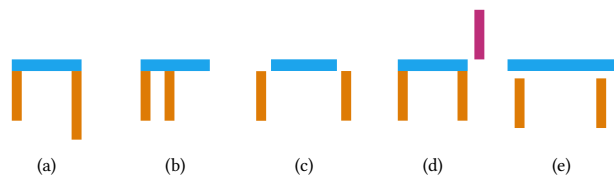


Fig. 7. Illustration of different cases addressed by refinement optimization: (a) equal length constraint, (b) stable support constraint, (c-e) support constraint.

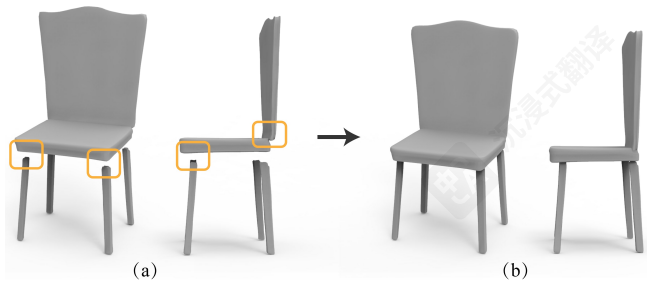


Fig. 8. An example showing the effect of shape refinement. (a) shape decoded by SDM-NET directly, (b) shape after optimization. The artifacts of gap and unstable support are fixed. The sub-figures show the same shape from two viewpoints, with problematic areas highlighted.

allowed and is set to $\varepsilon = 0.1$ in our experiments. 2) assuming $\tilde{b}_i$ and $\tilde{b}_j$ are the bounding boxes of parts i and j projected onto the plane orthogonal to the supporting direction t , it should satisfy that either $\tilde{b}_i \subseteq \tilde{b}_j$ or $\tilde{b}_j \subseteq \tilde{b}_i$ (see Figure 7 (c-d) for examples). This constraint can be formulated as an integer programming problem and solved efficiently during the optimization. The detailed formulation is provided in the Appendix.

Stable Support Constraint: For the “support above” relation ($t = 1$), the center of a supported part should be located in the supporting bounding box (the bounding box that covers all the bounding boxes of the supporting parts) for stable support (see Figure 7 (b) for an example violating this constraint). For a single supported part, the following constraints should be followed. $p'_i[l] - q'_i[l] \leq p'_j[l] \leq p'_i[l] + q'_i[l]$, $l \in \{0, 2\}$. For multiple supporting parts (e.g. four legs supporting the table top), the lower bound and upper bound of the x and z directions will be chosen from the corresponding parts.

This quadratic optimization with linear integer programming is solved by TOMLAB [Holmström and Edvall 2004] efficiently. We show an example of shape refinement in Figure 8.

4 DATASET AND NETWORK IMPLEMENTATION

We now give the details of our network architecture and training process. The experiments were carried out on a computer with an i7 6850K CPU, 64GB RAM, and a GTX 1080Ti GPU.

4.1 Dataset Preparation

The mesh models used in our paper are from [Yi et al. 2016], including a subset of ShapeNet Core V2 models [Chang et al. 2015], as well as ModelNet [Wu et al. 2015]. These datasets include pre-aligned models. However, ModelNet does not contain semantic segmentation, and models from [Yi et al. 2016] sometimes do not have sufficiently detailed segmentation to describe support structure (e.g. the car body and four wheels are treated as a single segment). To facilitate our processing, we use an active learning approach [Yi et al. 2016] to perform a refined semantic segmentation. We further use refined labels to represent individual parts of the same type, e.g., to have *left armrest* and *right armrest* labels for two armrest parts. The statistics of the resulting dataset are shown in Table 1.

Our network takes 3D shapes with consistent segmentation as input. The segmentation of test shapes can be obtained by some supervised methods such as [Qi et al. 2017,?]. Each segmented part is registered from the bounding box by non-rigid deformation. Our method allows each part type to include substantial geometric variations, e.g., the swivel leg and bar stool in Figure 14. Within the confines of the consistent segmentation, the SP-VAE is capable of handling variations of part structure and topologies, as well as varying part counts (by marking certain parts as non-existing).

Category	Airplane	Car	Chair	Table	Mug	Monitor	Guitar
# Meshes	2690	1824	3746	5266	213	465	787
# Labels	14	7	10	9	2	3	3

Table 1. The numbers of meshes and total part labels for each category in our dataset (after label refinement).

4.2 Network Architecture

The whole network includes two components, namely *PartVAE* for encoding the deformation of each part of a shape, and *SP-VAE* for jointly encoding the global structure of the shape and the geometric details of each part.

As illustrated in Figure 4, the structure of the PartVAE has two convolutional layers and one fully connected layer. We use *tanh* as the activation function, and in the last convolution layer, we use the linear output. The output of the last convolution layer is reshaped to a vector and mapped into a 64-dimensional latent space by the fully connected layer. The decoder has a mirrored structure, but not sharing weights with the encoder. We train the PartVAE once for each part type.

The input of the SP-VAE is the concatenated representation vector of all parts as shown in Figure 6. The input is fully connected with dimensions 1024, 512 and 256, respectively, and the latent space dimension is 128. Leaky ReLU is set as the activation function.

4.3 Parameters

We use fixed hyper-parameters in our experiments for different shape categories. In the following, we perform experiments on the *table* data in the ShapeNet Core V2 to demonstrate how the method behaves with changing hyper-parameters. The dataset is randomly

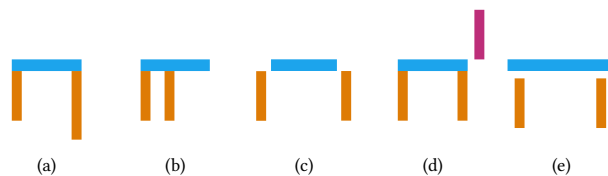


图7. 细化优化处理的不同案例说明: (a)等长约束, (b)稳定支撑约束, (c-e)支撑约束。

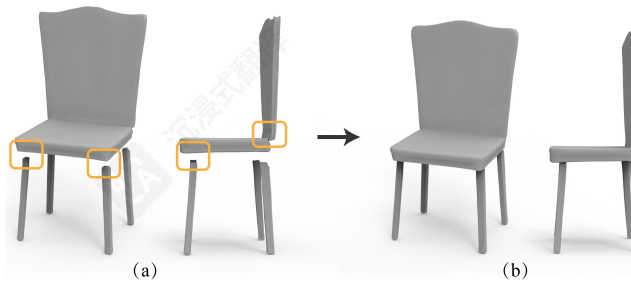


图8. 形状细化效果的示例。(a) SDM-NET直接解码的形状, (b)优化后的形状。间隙和不稳定支撑的伪影得到修复。子图从两个视角展示相同形状, 问题区域已高亮。

允许且在我们的实验中设置为 $\varepsilon = 0.1$ 。2) 假设 $\tilde{b}_i$ 和 $\tilde{b}_j$ 是部件 i 和 j 在垂直于支撑方向 t 的正交平面上的包围盒, 则应满足 $\tilde{b}_i \subseteq \tilde{b}_j$ 或 $\tilde{b}_j \subseteq \tilde{b}_i$ (见图7 (c-d)所示示例)。此约束可以表述为一个整数规划问题, 并在优化过程中高效求解。详细公式在附录中提供。

稳定支撑约束: 对于“支撑在上”关系 ($t = 1$), 被支撑部分的中心应位于支撑边界框 (覆盖所有支撑部分边界框的边界框) 内, 以实现稳定支撑 (见图7 (b)所示违反此约束的示例)。对于单个被支撑部分, 应遵循以下约束。 $p'_i[l] - q'_i[l] \leq p'_j[l] \leq p'_i[l] + q'_i[l]$, $l \in \{0, 2\}$ 。对于多个支撑部分 (例如四个腿支撑桌面), x 和 z 方向的下限和上限将从中选择相应的部分。

这种二次优化与线性整数规划问题由 TOMLAB [Holmström and Edvall 2004] 高效解决。我们在图8中展示了一个形状优化的示例。

4 数据集和网络实现

我们现将网络架构和训练过程的详细信息提供如下。实验是在一台配备 i7 6850K CPU、64GB 内存和 GTX 1080Ti GPU 的计算机上进行的。

4.1 数据集准备

我们论文中使用的网格模型来自 [Yi等人 2016], 包括 ShapeNet Core V2模型 [Chang等人 2015]的子集以及 ModelNet [Wu等人 2015]。这些数据集包含预对齐的模型。然而, ModelNet 不包含语义分割, 而来自 [Yi等人 2016] 的模型有时分割细节不足, 无法描述支撑结构 (例如, 车身和四个轮子被视为一个单一部分割)。为了便于处理, 我们采用主动学习方法 [Yi等人 2016] 进行精细的语义分割。我们进一步使用精细标签来表示同类部件的各个部分, 例如, 为两个扶手部件分别设置左扶手和右扶手标签。最终数据集的统计信息显示在表1中。

我们的网络以具有一致分割的3D形状为输入。测试形状的分割可以通过一些监督方法 (如 [Qi等人, 2017, ?]) 获得。每个分割部分通过非刚性变形从边界框中注册。我们的方法允许每种部分类型包含显著的几何变化, 例如图14中的旋转腿和吧台椅。在一致分割的范围内, SP-VAE能够处理部分结构和拓扑的变化, 以及部分数量的变化 (通过将某些部分标记为不存在)。

类别	飞机	汽车	椅子	桌子	杯子	显示器	吉他
# 网格	2690	1824	3746	5266	213	465	787
# 标签	14	7	10	9	2	3	3

表1. 我们数据集中每个类别的网格数量和总标签数 (标签优化后)。

4.2 网络架构

整个网络包含两个组件, 分别是 PartVAE 用于编码形状各部分的变形, 以及 SP-VAE 用于联合编码形状的全局结构和各部分的几何细节。

如图4所示, PartVAE 的结构包含两层卷积层和一层全连接层。我们使用 tanh 作为激活函数, 并在最后一个卷积层使用线性输出。最后一个卷积层的输出被重塑为一个向量, 并通过全连接层映射到64维的潜在空间。解码器具有镜像结构, 但不与编码器共享权重。我们对每种部分类型训练一次 PartVAE。

SP-VAE 的输入是如图6所示的所有部分的连接表示向量。输入与维度分别为1024、512和256的全连接层完全连接, 潜在空间维度为128。Leaky ReLU 被设置为激活函数。

4.3 参数

我们在实验中针对不同的形状类别使用固定的超参数。在下文中, 我们在 ShapeNet CoreV2 的表格数据上执行实验, 以演示该方法在超参数变化时的表现。该数据集被随机

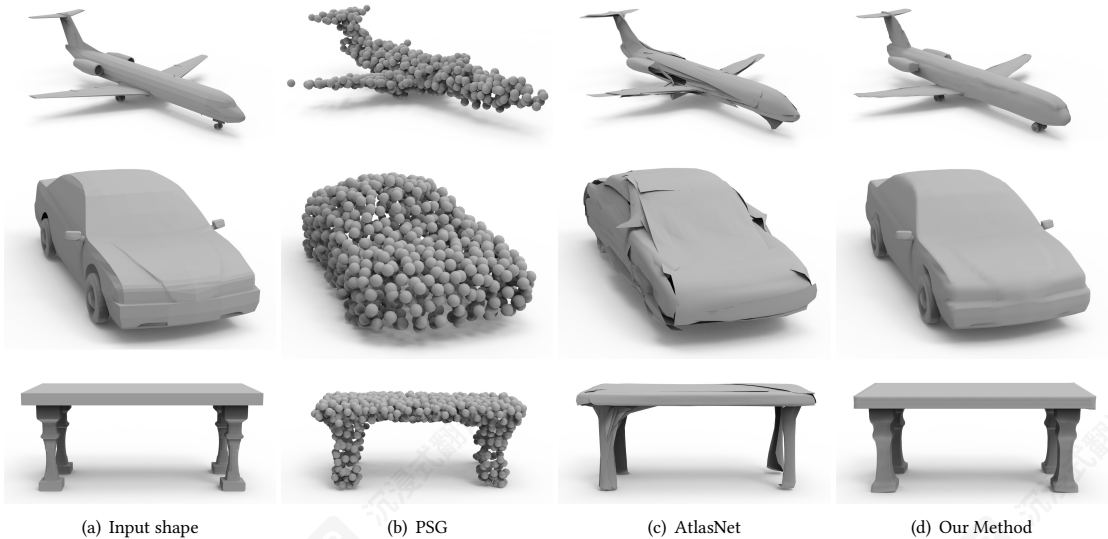


Fig. 9. Representative results of decoded shapes with different methods. Compared with PSG [Fan et al. 2017] and AtlasNet [Groueix et al. 2018], our method produces the decoded shapes of higher quality. PSG results are rather coarse point samples. The results by AtlasNet exhibit clearly noticeable patch artifacts.

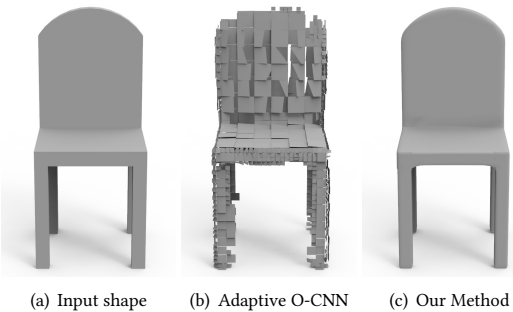


Fig. 10. Visual comparison of the decoded shapes with Adaptive O-CNN [Wang et al. 2018b] and our method. Compared with Adaptive O-CNN, while the planar regions of the chair can be decoded by both methods, the curved regions such as the top of the chair back can be recovered only by our method.

split into the training data (75%) and test data (25%). The generalization of SP-VAE is evaluated with different hyper-parameters in Table 2, where the bidirectional Chamfer distance is used to measure the reconstruction error on the test data (as unseen data). We perform such tests for 10 times and report the average errors in Table 2. As can be seen, SP-VAE has the lowest error with the hyper-parameters $\lambda_1 = 1.0$ and $\lambda_2 = 0.5$, where λ_1 and λ_2 are the weights of the reconstruction error term and KL divergence term, respectively. The hyper-parameters (weights of reconstruction, KL-divergence, and regularization) of PartVAE are set to the same numbers in [Gao et al. 2018]. We set the dimension of the latent space of PartVAE to 64, and the dimension of the latent space of SP-VAE to 128. These two parameters are evaluated in Tables 3

and 4 with the reconstruction error. When adjusting the dimension of one VAE, we leave the dimension of the other VAE unchanged.

(λ_1, λ_2)	(0.5, 0.5)	(1.0, 0.5)	(0.5, 1.0)	(1.0, 1.0)
Recons. Error ($\times 10^{-3}$)	2.01	1.85	2.24	1.94

Table 2. Comparison of average SP-VAE reconstruction errors (measured in bidirectional Chamfer distance) for unseen data on the *table* dataset w.r.t. changing hyper-parameters.

4.4 Training Details

Since a PartVAE encodes the geometry of a specific type of parts, it is trained separately. SP-VAE is then trained using PartVAE for encoding part geometry. Training of both VAEs is optimized using the Adam solver [Kingma and Ba 2015]. The PartVAE is trained with 20,000 iterations and SP-VAE with 120,000 iterations by minimizing their loss functions. For both VAEs, we set the batch size as 512 and learning rate starting from 0.001 and decaying every 1000 steps with the decay rate set to 0.8. The training batch is randomly sampled from the training data set.

For a typical category, the training of both PartVAE and SP-VAE takes about 300 minutes. Once the networks are trained, shape

PartVAE Embedding Dimension	32	64	128	256
PartVAE Recons. Error ($\times 10^{-3}$)	1.92	1.76	1.74	1.82
SP-VAE Recons. Error ($\times 10^{-3}$)	2.16	1.85	1.91	2.03

Table 3. Comparison of average reconstruction errors (measured by bidirectional Chamfer distance) for PartVAE and SP-VAE w.r.t. changing dimension of the PartVAE latent space.

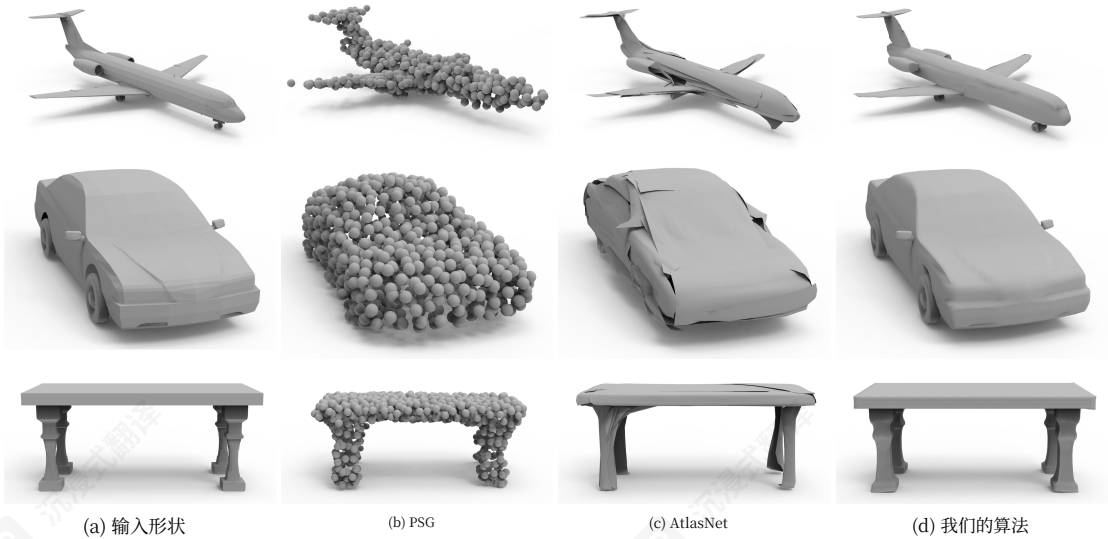


图9. 不同方法解码形状的代表结果。与PSG [Fan等人 2017] 和AtlasNet [Groueix等人 2018], 相比, 我们的方法产生了更高质量的解码形状。PSG结果相当粗糙的点采样。AtlasNet的结果表现出明显的块状伪影。

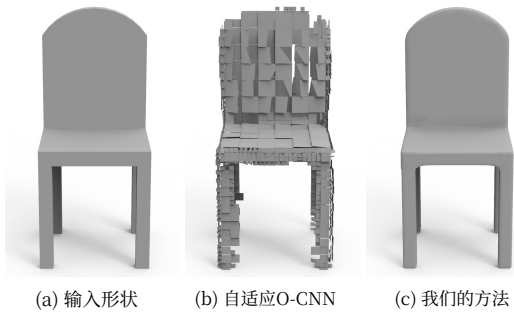


图10. Adaptive O-CNN [Wang等人2018b]和我们的方法解码形状的视觉比较。与自适应O-CNN相比, 虽然椅子的平面区域两种方法都能解码, 但椅背顶部等曲面区域只能通过我们的方法恢复。

分为训练数据 (75%) 和测试数据 (25%)。SP-VAE 的泛化能力在表 2 中使用不同的超参数进行评估, 其中双向 Chamfer 距离用于测量测试数据 (作为未见数据) 上的重建误差。我们进行了 10 次此类测试, 并在表 2 中报告了平均误差。可以看出, SP-VAE 在超参数 $\lambda_1 = 1.0$ 和 $\lambda_2 = 0.5$ 下具有最低误差, 其中 λ_1 和 λ_2 分别是重建误差项和 KL 散度的权重。PartVAE 的超参数 (重建、KL 散度和平滑项的权重) 设置为 [Gao et al. 2018] 中的相同数值。我们将 PartVAE 的潜在空间维度设置为 64, SP-VAE 的潜在空间维度设置为 128。这两个参数在表 3

以及重建误差。当调整一个VAE的维度时, 我们保持另一个VAE的维度不变。

(λ_1, λ_2)	(0.5, 0.5)	(1.0, 0.5)	(0.5, 1.0)	(1.0, 1.0)
重建误差 ($\times 10^{-3}$)	2.01	1.85	2.24	1.94

表2. 在表数据集上, 针对未见数据, 比较平均SP-VAE重建误差 (以双向Chamfer距离衡量) 随超参数变化的情况。

4.4 训练细节

由于PartVAE编码特定类型零件的几何形状, 因此它需要单独训练。SP-VAE随后使用PartVAE进行编码几何形状的训练。这两个VAE的训练都使用Adam求解器 [Kingma和 2015] Ba进行优化。PartVAE使用20,000次迭代, SP-VAE使用120,000次迭代, 通过最小化它们的损失函数进行训练。对于这两个VAE, 我们设置批大小为512, 学习率从0.001开始, 每1000步衰减一次, 衰减率设置为0.8。训练批次从训练数据集中随机采样。

对于一个典型类别, PartVAE和SP-VAE的训练大约需要300分钟。一旦网络训练完成, 形状

PartVAE嵌入维度	32	64	128	256
PartVAE重建误差 ($\times 10^{-3}$)	1.92	1.76	1.74	1.82
SP-VAE重建误差 ($\times 10^{-3}$)	2.16	1.85	1.91	2.03

表3. PartVAE和SP-VAE在不同PartVAE潜在空间维度下的平均重建误差比较 (以双向Chamfer距离衡量)。

SP-VAE Embedding Dimension	32	64	128	256
SP-VAE Recons. Error ($\times 10^{-3}$)	2.23	1.99	1.85	1.91

Table 4. Comparison of average reconstruction errors (measured by bidirectional Chamfer distance) of SP-VAE w.r.t. changing embedding dimension.

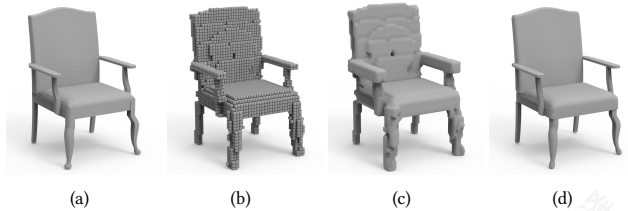


Fig. 11. Visual comparison of shape reconstruction with GRASS [Li et al. 2017] and our technique. (a) input shape, (b)(c) GRASS results in voxels and extracted mesh, (d) Our result.

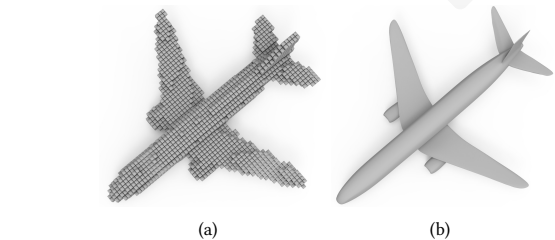


Fig. 12. Visual comparison between the global-to-local method [Wang et al. 2018a] (a) and our technique (b) for shape generation.

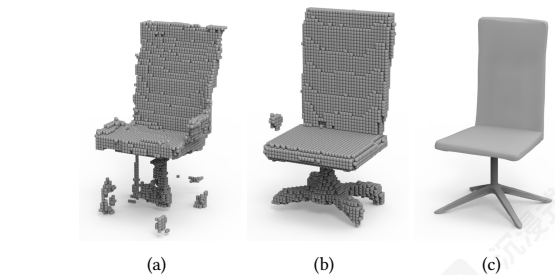


Fig. 13. Visual comparison of shape generation using different methods. (a) 3DGAN [Wu et al. 2016], (b) the global-to-local method [Wang et al. 2018a], (c) our technique.

generation is very efficient: generating one shape and structure optimization take only about 36 and 100 milliseconds, respectively.

5 RESULTS AND EVALUATION

We present the results of shape reconstruction, shape generation and shape interpolation to demonstrate the capability of our method, and compare them with those generated by the state-of-the-art methods. We also perform ablation studies to show the advantages of our

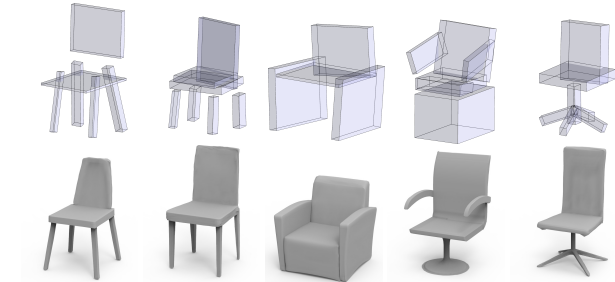


Fig. 14. Comparison between GRASS [Li et al. 2017] and our method for random generation. We visualize the structures generated by GRASS and shapes generated by our method.

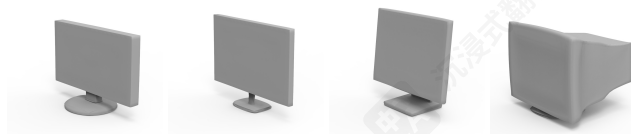


Fig. 15. Random generation of monitor shapes using our method, where the training data is from ModelNet [Wu et al. 2015].

design. Finally, we present examples to show generalizability (i.e., applying our learned model to new shapes of the same category), editability and limitations of our technique.

Shape Reconstruction. We compare our method with PSG [Fan et al. 2017], AtlasNet [Groueix et al. 2018] and Adaptive O-CNN [Wang et al. 2018b] on the ShapeNet Core V2 dataset. In this experiment, we choose four representative categories commonly used in the literature to perform both qualitative and quantitative comparisons. Each dataset is randomly split into the training set (75%) and test set (25%). For fair comparison, we train PSG, AtlasNet, and Adaptive O-CNN for individual shape categories, similar to ours. To prepare the input for PSG, we use rendered images under different viewpoints. Given the same input models in the test set, we compare the decoded shapes by different methods. Figures 9 and 10 show the visual comparison of representative results on several test shapes. It can be easily seen that the decoded shapes by PSG, Adaptive O-CNN and AtlasNet cannot capture the shapes faithfully. AtlasNet and Adaptive O-CNN are able to produce more details than PSG, but suffer from clearly noticeable patch artifacts. In contrast, SDM-NET recovers shapes with higher quality and finer-detailed geometry. Note that we compare the existing methods with SP-VAE followed by structure optimization instead of SP-VAE alone, since structure optimization, which is dependent on the output of SP-VAE, is a unique and essential component in our system, and cannot be directly used with the methods being compared due to their lack of structure information.

Moreover, we quantitatively compare our method with the existing methods using common metrics for 3D shape sets, including Jensen-Shannon Divergence (JSD), Coverage (COV) and Minimum Matching Distance (MMD) [Achlioptas et al. 2018]. The latter two metrics are calculated using both the Chamfer Distance (CD) and Earth Mover’s Distance (EMD) for measuring the distance between

SP-VAE 嵌入维度	32	64	128	256
SP-VAE 重建误差 ($\times 10^{-3}$)	2.23	1.99	1.85	1.91

表4. SP-VAE 在嵌入维度变化时平均重建误差的比较（通过双向 Chamfer距离衡量）。

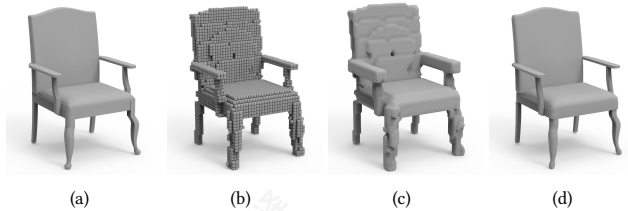


图11. 形状重建的可视化比较: GRASS [Li 等人 2017] 与我们的技术。(a) 输入形状, (b)(c) GRASS 在体素和提取网格中的结果, (d) 我们的结果。

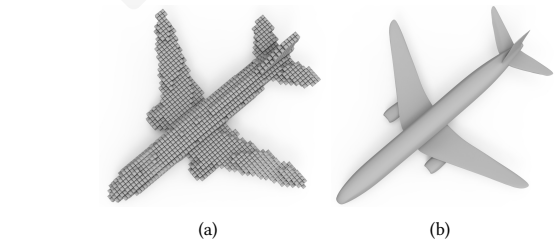


图12. 全局到局部方法 [Wangetal. 2018a] (a) 与我们的技术 (b) 在形状生成方面的视觉比较。

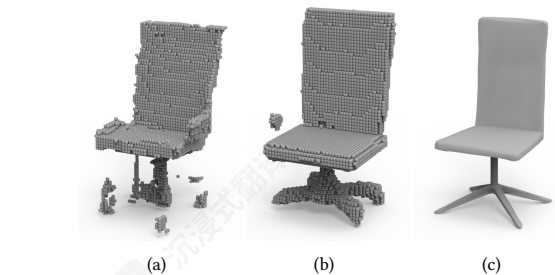


图13. 使用不同方法的形状生成视觉比较。(a) 3DGAN [Wu 等人 2016], (b) 全局到局部方法 [Wang 等人 2018a], (c) 我们的技术。

生成非常高效: 生成一个形状和结构优化分别只需要大约 36 和 100 毫秒。

5 结果与评估

我们展示了形状重建、形状生成和形状插值的结果, 以展示我们方法的能力, 并将它们与最先进方法生成的结果进行比较。我们还进行了消融研究, 以展示我们设计的优势。

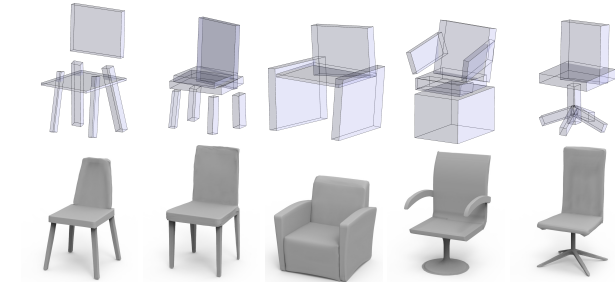


图14. GRASS [Li 等人 2017] 与我们方法在随机生成方面的比较。我们可视化 GRASS 生成的结构和我们的方法生成的形状。

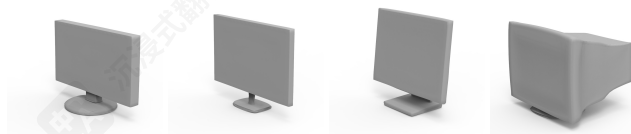


图15. 使用我们的方法进行监控形状的随机生成, 其中训练数据来自 ModelNet [Wu et al. 2015].

最后, 我们展示了示例, 以展示我们技术的泛化能力(即, 将我们学习的模型应用于同一类别的新的形状)、可编辑性和局限性。

形状重建。我们将我们的方法与 PSG [Fan et al. 2017], AtlasNet [Groueix et al. 2018] 以及 Adaptive O-CNN [Wang et al. 2018b] 在 ShapeNet Core V2 数据集上进行比较。在这个实验中, 我们选择文献中常用的四个代表性类别进行定性和定量比较。每个数据集随机分为训练集 (75%) 和测试集 (25%)。为了公平比较, 我们为每个形状类别单独训练 PSG、AtlasNet 和 Adaptive O-CNN, 类似于我们的方法。为了准备 PSG 的输入, 我们使用在不同视点下的渲染图像。在测试集中给出相同的输入模型, 我们比较不同方法解码的形状。图9和图10显示了几个测试形状上代表性结果的视觉比较。可以看出, PSG、Adaptive O-CNN 和 AtlasNet 解码的形状无法忠实捕捉形状。AtlasNet 和 Adaptive O-CNN 能够产生比 PSG 更多的细节, 但存在明显的块状伪影。相比之下, SDM-NET 恢复了更高质量和更精细几何细节的形状。请注意, 我们比较现有方法时使用的是 SP-VAE 后跟结构优化, 而不是单独的 SP-VAE, 因为结构优化依赖于 SP-VAE 的输出, 是我们系统中独特且重要的组成部分, 而由于比较方法缺乏结构信息, 无法直接使用。

此外, 我们使用 Jensen-Shannon 散度 (JSD)、覆盖率 (COV) 和最小匹配距离 (MMD) 等常用指标, 定量比较了我们的方法与现有方法, [Achlioptas et al. 2018]。后两种指标在计算时均使用 Chamfer 距离 (CD) 和 Earth Mover’s 距离 (EMD) 来测量形状之间的距离



Fig. 16. Visual comparison of shape interpolation by AtlasNet [Groueix et al. 2018] (top) and our technique (bottom). The first and last columns are the two shapes to be interpolated (which are decoded by respective methods).



Fig. 17. Shape interpolation of cups and chairs with different topologies using our method. The first and last columns are the input shapes for interpolation.

shapes. For JSD and MMD, the smaller the better, while for COV, the larger the better. The average results for different methods on these datasets are shown in Table 5. It can be seen that our method achieves the best performance for nearly all the metrics.

In Table 6 we compare our method with three recent shape generation methods, namely, GRASS [Li et al. 2017], G2L [Wang et al. 2018a] and SAGNet [Wu et al. 2019]. For fair comparison, both of our reconstructed shapes and input shapes are voxelized. Particularly, we make comparisons with GRASS on their chair data since GRASS requires symmetry hierarchies as input for training. The results show that our method outperforms the compared methods in most cases with several metrics. We also show a visual comparison result between GRASS and our method in Figure 11. The GRASS result exhibits some artifacts due to the voxel representation. Even after surface extraction GRASS still fails to capture fine geometric details compared with our method.

Shape Generation. In Figure 12, we make a qualitative comparison between our technique and the global-to-local method [Wang et al. 2018a] by randomly generating shapes of airplanes. Their method uses an unconditional GAN architecture and thus cannot reconstruct a specific shape. So two randomly generated, visually similar planes are selected for comparisons. Their voxel based method fails to represent smooth, fine details of 3D shapes. We make further comparison with the global-to-local method [Wang et al. 2018a] as well as 3DGAN [Wu et al. 2016] in Figure 13. Again, we select visually similar shapes for comparison, and our method produces high quality shapes with plausible structure and fine details, whereas alternative methods have clear artifacts including fragmented output and rough surfaces. We also compare our technique with GRASS [Li et al. 2017] for random shape generation. As shown in Figure 14, the structures synthesized by GRASS might be problematic, containing parts which are disjoint and/or not well supported. In addition,

since it is trained on symmetry hierarchies constructed on top of the shape segmentations, once trained, for new inputs GRASS utilizes automatically generated symmetry hierarchies, which, however, can be inconsistent. This is one of the main causes for GRASS to produce results with a greater level of structural noise including disconnections and asymmetries. In contrast, our results are physically stable and well connected. Note that our refinement step is an integral part of our pipeline and requires structure relations, so cannot be directly applied to GRASS.

As a generative model, our technique is able to generate new shapes. Because our architecture consists of two VAEs, i.e., PartVAE and SP-VAE, we can acquire different information from their latent spaces. Specifically, we extract various types of parts and structural information from the latent space of SP-VAE, and combine them with the deformation information from PartVAE, to produce novel shapes. Figure 15 gives an example, where our method is used to generate computer monitors with various shapes by sampling in the learned latent space. In this example, the training data is obtained from ModelNet [Wu et al. 2015].

Shape Interpolation. Shape interpolation is a useful technique to generate gradually changing shape sequences between a source shape and a target shape. With the help of SP-VAE, we first encode the source and target shapes into latent vectors and then perform linear interpolation in the latent space of VAE. A sequence of shapes between the input shape pairs are finally decoded from the linearly interpolated latent vectors. In Figure 16, we compare our technique with AtlasNet [Groueix et al. 2018] for their performance on shape interpolation. It can be easily seen that the results by AtlasNet suffer from patch artifacts and the surfaces of the interpolated shapes are often not very smooth. The interpolation in our latent space leads to much more realistic results. For example, the armrests gradually become thinner and then disappear in a more natural manner. This is because we combine the geometry and structure during the training of SDM-NET, which thus learns the implicit joint distribution of the geometry and structure.

The effectiveness of shape interpolation with SDM-NET is consistently observed with additional experiments on different datasets. Figure 17 shows two examples of natural interpolation between shapes with different topologies, thanks to our flexible structure representation. Figure 18 shows an additional interpolation result with substantial change of geometry.

Ablation Studies. We perform several ablation studies to demonstrate the necessity of key components of our architecture.

Support vs. adjacency relationships. We adopt support relationships in our method, rather than adjacency relationships to get well connected shapes, because support relationships ensure generating physically stable shapes, and provide a natural order which is useful to simplify the structure refinement optimization. In contrast, using a bidirectional adjacency, it would be much more complicated to formulate and optimize constraints between two adjacent parts. To evaluate the effectiveness of the support relationships, we replace the support constraints by simply minimizing the distance between every pair of adjacent parts to approximate the adjacency relationships. The effects of using support and adjacency constraints are



图16. AtlasNet [Groueix等人2018] (顶部)和我们的技术(底部)的形状插值视觉比较。第一列和最后一列是待插值的两个形状(分别由相应方法解码)。



图17. 使用我们的方法对不同拓扑结构的杯子和椅子进行形状插值。第一列和最后一列是插值的输入形状。

。对于 JSD 和 MMD, 数值越小越好, 而 COV 则越大越好。表 5 展示了不同方法在这些数据集上的平均结果。可以看出, 我们的方法在几乎所有指标上均取得了最佳性能。

在表6中, 我们将我们的方法与三种最近的形状生成方法进行比较, 即GRASS [Li等人 2017], G2L [Wang等人2018a] 和 SAGNet [Wu et al. 2019]。为了公平比较, 我们重建的形状和输入形状都进行了体素化。特别地, 我们在他们的椅子数据上与 GRASS进行了比较, 因为GRASS需要对称层次结构作为训练输入。结果表明, 我们的方法在大多数情况下优于比较方法, 在几个指标上表现更好。我们还展示了GRASS和我们的方法之间的视觉比较结果(图11)。由于体素表示, GRASS结果表现出一些伪影。即使进行表面提取, GRASS与我们的方法相比仍然无法捕捉精细的几何细节。

形状生成。在图12中, 我们通过随机生成飞机形状, 对我们的技术与全局到局部方法 [Wang等人2018a] 进行了定性比较。他们的方法使用无条件的GAN架构, 因此无法重建特定形状。因此, 选择了两个随机生成的、视觉上相似的飞机进行比较。他们的基于体素的方法无法表示3D形状的平滑、精细细节。我们进一步与全局到局部方法 [Wang等人2018a] 以及 3DGAN [Wu等人 2016] 在图13中进行了比较。同样, 我们选择了视觉上相似的形状进行比较, 我们的方法产生了具有合理结构和精细细节的高质量形状, 而其他方法存在明显的伪影, 包括碎片化输出和粗糙表面。我们还将我们的技术与GRASS [Li等人 2017] 用于随机形状生成进行了比较。如图14所示, GRASS合成的结构可能存在问题, 包含部分是分离的或/且支撑不佳。此外,

由于GRASS是在基于形状分割构建的对称层次结构上训练的, 一旦训练完成, 对于新输入, GRASS会利用自动生成的对称层次结构, 然而, 这些层次结构可能不一致。这是GRASS产生具有更高结构噪声结果的主要原因, 包括断开和不对称性。相比之下, 我们的结果在物理上是稳定的且连接良好。请注意, 我们的细化步骤是我们流程的有机组成部分, 需要结构关系, 因此不能直接应用于GRASS。

作为一种生成模型, 我们的技术能够生成新的形状。由于我们的架构由两个VAE组成, 即PartVAE和SP-VAE, 我们可以从它们的潜在空间中获取不同的信息。具体来说, 我们从 SP-VAE的潜在空间中提取各种类型的部件和结构信息, 并将它们与来自PartVAE的变形信息结合起来, 以生成新的形状。图15给出了一个示例, 其中我们的方法用于通过在学习的潜在空间中进行采样来生成各种形状的计算机显示器。在这个示例中, 训练数据来自ModelNet [Wu等人 2015]。

形状插值。形状插值是一种有用的技术, 用于在源形状和目标形状之间生成逐渐变化的形状序列。借助SP-VAE, 我们首先将源形状和目标形状编码为潜在向量, 然后在VAE的潜在空间中进行线性插值。最终, 从线性插值的潜在向量中解码出输入形状对之间的形状序列。在图16中, 我们将我们的技术与 AtlasNet [Groueix等人 2018] 在形状插值方面的性能进行了比较。可以看出, AtlasNet的结果存在块状伪影, 插值形状的表面通常不够平滑。我们的潜在空间插值产生了更逼真的结果。例如, 扶手逐渐变薄, 然后以更自然的方式消失。这是因为我们在SDM-NET的训练过程中结合了几何和结构, 因此它学习了几何和结构的隐式联合分布。

通过在不同数据集上进行额外实验, 观察到SDM-NET形状插值的有效性。图17展示了两种具有不同拓扑结构的形状之间进行自然插值的示例, 这得益于我们灵活的结构表示。图 18展示了一个几何形状发生显著变化的额外插值结果。

消融实验。我们进行了多项消融实验, 以证明我们架构中关键组件的必要性。

支持与邻接关系。在我们的方法中, 我们采用支持关系而不是邻接关系来获得连接良好的形状, 因为支持关系确保生成物理稳定的形状, 并提供一个有用的自然顺序, 有助于简化结构细化优化。相比之下, 使用双向邻接, 在制定和优化两个相邻部分之间的约束将会复杂得多。为了评估支持关系的效果, 我们通过简单地最小化每对相邻部分之间的距离来替换支持约束, 以近似邻接关系。使用支持和邻接约束的效果是

Dataset	Methods	Metrics				
		JSD	MMD-CD	MMD-EMD	COV-CD	COV-EMD
Airplane	AOCNN	0.0665	0.0167	0.0157	84.3	95.5
	AtlasNet	0.0379	0.0147	0.0132	79.6	82.1
	PSG	0.0681	0.0244	0.0172	33.5	38.9
	Our	0.0192	0.00462	0.00762	87.2	90.6
Car	AOCNN	0.0649	0.0264	0.0223	60.6	60.8
	AtlasNet	0.0393	0.0228	0.0137	75.4	81.9
	PSG	0.0665	0.0365	0.0247	49.8	59.4
	Our	0.0280	0.00247	0.00101	87.2	88.5
Chair	AOCNN	0.0384	0.0159	0.0196	43.5	39.3
	AtlasNet	0.0369	0.0137	0.0124	51.1	52.6
	PSG	0.0391	0.0131	0.0152	42.9	49.1
	Our	0.0364	0.00375	0.00764	47.3	55.3
Table	AOCNN	0.0583	0.0393	0.0256	55.2	40.1
	AtlasNet	0.0324	0.0154	0.0146	59.1	63.7
	PSG	0.0354	0.0271	0.0276	41.2	42.5
	Our	0.0123	0.00183	0.00127	63.3	76.8

Table 5. Quantitative comparison of reconstruction capabilities of different methods on several metrics. For JSD and MMD, the smaller the better, while for COV, the larger the better.

Dataset	Methods	Metrics				
		JSD	MMD-CD	MMD-EMD	COV-CD	COV-EMD
Chair	G2L	0.0357	0.0034	0.0682	83.7	83.4
	GRASS	0.0374	0.0030	0.0744	46.0	44.5
	SAGNet	0.0342	0.0024	0.0608	75.1	74.3
	Our	0.0289	0.00274	0.00671	89.3	84.1

Table 6. Quantitative comparison of reconstruction capabilities of different methods (G2L [Wang et al. 2018a], GRASS [Li et al. 2017], SAGNet [Wu et al. 2019]) on several metrics. For JSD and MMD, the smaller the better, while for COV, the larger the better.

Dataset	Car	Chair	Guitar	Airplane	Table
Separate	2.77	3.89	3.58	4.87	1.85
End-to-End	5.07	6.73	7.44	11.38	4.86

Table 7. Comparison of reconstruction errors ($\times 10^{-3}$) under the metric of bidirectional Chamfer distance with two different training strategies, i.e., separate training vs. end-to-end training.

shown in Figure 19. It can be seen that the support constraints lead to a physically more stable result.

Separate vs. end-to-end training. We adopt separate training for the two-level VAE, i.e. PartVAEs are trained for individual part types first, before training SP-VAE where the geometries of parts are encoded with the trained PartVAEs. The two-level VAE could

数据集	方法	指标				
		JSD	MMD-CD	MMD-EMD	COV-CD	COV-EMD
飞机	AOCNN	0.0665	0.0167	0.0157	84.3	95.5
	AtlasNet	0.0379	0.0147	0.0132	79.6	82.1
	PSG	0.0681	0.0244	0.0172	33.5	38.9
	Our	0.0192	0.00462	0.00762	87.2	90.6
Car	AOCNN	0.0649	0.0264	0.0223	60.6	60.8
	AtlasNet	0.0393	0.0228	0.0137	75.4	81.9
	PSG	0.0665	0.0365	0.0247	49.8	59.4
	Our	0.0280	0.00247	0.00101	87.2	88.5
Chair	AOCNN	0.0384	0.0159	0.0196	43.5	39.3
	AtlasNet	0.0369	0.0137	0.0124	51.1	52.6
	PSG	0.0391	0.0131	0.0152	42.9	49.1
	Our	0.0364	0.00375	0.00764	47.3	55.3
Table	AOCNN	0.0583	0.0393	0.0256	55.2	40.1
	AtlasNet	0.0324	0.0154	0.0146	59.1	63.7
	PSG	0.0354	0.0271	0.0276	41.2	42.5
	Our	0.0123	0.00183	0.00127	63.3	76.8

表 5. 不同方法在多个指标上的重建能力定量比较。对于 JSD 和 MMD，越小越好，而 COV 越大越好。

数据集	方法	指标				
		JSD	MMD-CD	MMD-EMD	COV-CD	COV-EMD
椅子	G2L	0.0357	0.0034	0.0682	83.7	83.4
	草地	0.0374	0.0030	0.0744	46.0	44.5
	SAGNet	0.0342	0.0024	0.0608	75.1	74.3
	Our	0.0289	0.00274	0.00671	89.3	84.1

表6. 不同方法在多个指标上的重建能力定量比较 (G2L [Wang等人 2018a], GRASS [Li等人 2017], SAGNet [Wu等人 2019])。对于JSD和MMD，越小越好，而COV则越大越好。

数据集	Car	椅子	吉他	飞机	表格
分离	2.77	3.89	3.58	4.87	1.85
端到端	5.07	6.73	7.44	11.38	4.86

表7. 在双向Chamfer距离指标下，比较两种不同训练策略（即分离训练与端到端训练）的重建误差($\times 10^{-3}$)。

如图19所示。可以看出，支撑约束导致了一个在物理上更稳定的结果。

分离式与端到端训练。我们采用两阶段VAE的分离式训练，即先为单个部件类型训练PartVAEs，然后再训练SP-VAE，在SP-VAE中部件的几何信息由已训练的PartVAEs编码。这种两阶段VAE可以

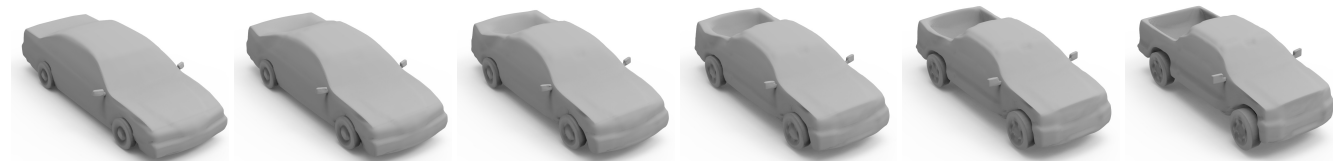


Fig. 18. Interpolating cars with different geometries using our method. The first and last columns are the shapes to be interpolated. The other columns are the in-between models by linear interpolation in the latent space.

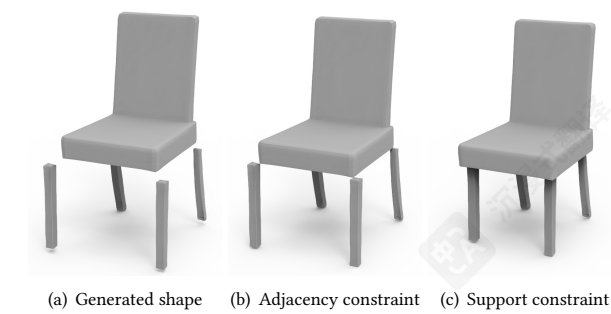


Fig. 19. Comparison between adjacency relationship constraint and support relationship constraint. (a) is a randomly generated extreme case by SDM-NET before refinement. (b) and (c) are the results after refinement with only the adjacency constraint and the support constraint, respectively.

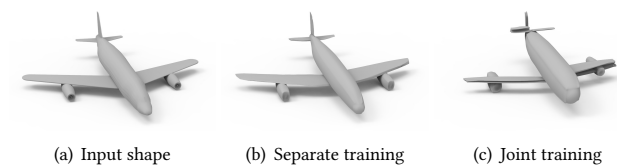


Fig. 20. Qualitative comparison between two different training strategies, i.e., separate training vs. end-to-end training. It can be seen that separate training keeps more geometric details.

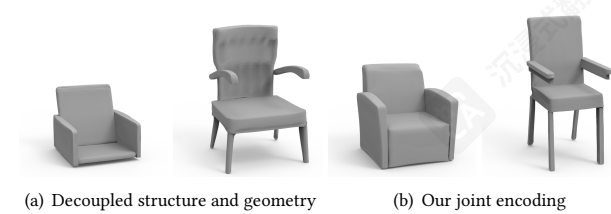


Fig. 21. Results of our method (b), compared with decoupled structure and geometry (a). The latter is produced by decoupling the geometry information from SP-VAE, and generating the geometry of individual parts independently.

also be trained end-to-end, i.e., optimizing both PartVAEs and SP-VAE simultaneously. We compare the average bidirectional Chamfer distance of the reconstruction of each part between end-to-end

training and separate training adopted in our solution, as given in Table 7. The visual comparisons are shown in Figure 20. Since without the help of the well-trained distribution of the latent space of individual parts, end-to-end training would result in optimization stuck at a poor local minimum, leading to higher reconstruction errors and visually poor results.

Joint vs. decoupled structure and part geometry encoding. In this paper, the geometry details represented as part deformations are encoded into the SP-VAE embedding space jointly (see Section 3.4). This approach ensures that generated shapes have consistent structure and geometry, and the geometries of different parts are also coherent. We compare our solution with an alternative approach where the structure and geometry encodings are decoupled: SP-VAE only encodes the structure and the geometry of each part is separately encoded using a PartVAE. Figure 21 shows randomly generated shapes with both approaches. The structure of the first example implies the shape is a sofa, but the geometry of the seat part in (a) does not look like a sofa, whereas our method generates part geometry consistent with the structure. For the second example, our method produces parts with coherent geometry (b), whereas using decoupled structure and geometry leads to inconsistent part styles (a).

Resolution of bounding boxes. By default, our method uses bounding boxes each with 19.2K triangles. We also try using lower and higher resolution bounding boxes. As shown in Figure 22, using lower resolution (b) cannot capture the details of the shape, and using higher resolution (d) produces very similar result as our default setting (c), but takes longer time. Our default setting (c) provides a good balance between efficiency and quality.

PartVAE per part type vs. single PartVAE. In our paper, we train a PartVAE for each part type. We compare this with an alternative approach where a single PartVAE is trained for all part categories. As shown in Figure 22 (e), this approach is not effective in capturing unique geometric features of different parts, leading to poor geometric reconstruction.

Generalizability. Figure 23 shows an example that demonstrates the generalizability of our method, to process new shapes of the same category without input semantic segmentation. We first train PointNet++ [Qi et al. 2017] on our labeled dataset, which is then used for semantic segmentation of the new shape. Finally, we obtain the reconstruction result by our SDM-NET. An example is shown in Figure 23, which demonstrates that semantic segmentation obtained automatically can be effectively used as input to our method.

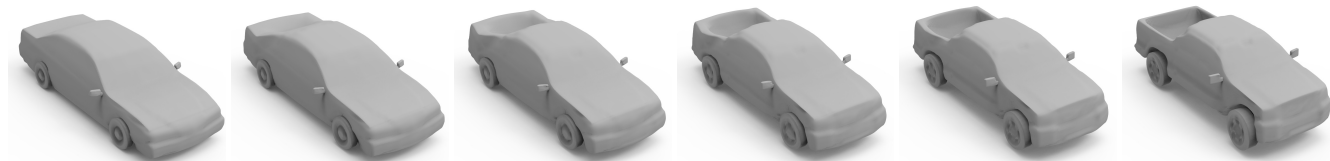


图18. 使用我们的方法对具有不同几何形状的插值汽车。第一列和最后一列是要插值的形状。其他列是潜在空间中线性插值的中间模型。



图19. 邻接关系约束与支持关系约束的比较。(a)是SDM-NET在细化之前随机生成的极端情况。(b)和(c)分别是仅使用邻接约束和仅使用支持约束细化后的结果。



图20. 两种不同训练策略（即分离训练与端到端训练）的定性比较。可以看出，分离训练保留了更多几何细节。

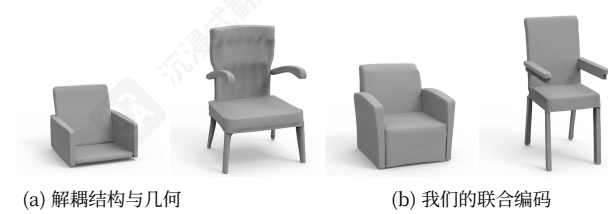


图21. 我们的方法 (b) 的结果，与解耦结构与几何 (a) 进行比较。后者是通过将几何信息从SP-VAE中解耦，并独立生成各个部分的几何形状产生的。

也可以进行端到端训练，即同时优化 PartVAEs 和 SP-VAE。我们比较每个部件重建的平均双向 Chamfer 距离

我们解决方案中采用的训练和独立训练方法，如表7所示。可视化比较结果展示在图20中。由于缺乏对个体部分潜在空间良好训练分布的帮助，端到端训练会导致优化陷入较差的局部最小值，从而造成更高的重建误差和视觉效果不佳的结果。

联合与解耦结构及部分几何编码。在本文中，表示为部分变形的几何细节被联合编码到SP-VAE嵌入空间中（参见第3.4节）。这种方法确保生成的形状具有一致的结构和几何特征，不同部分的几何特征也保持协调。我们将我们的解决方案与一种结构及几何编码解耦的替代方法进行比较：SP-VAE仅编码结构，而每个部分的几何特征则使用PartVAE单独编码。图21展示了两种方法随机生成的形状。第一个示例的结构暗示形状是沙发，但(a)中座位的几何形状并不像沙发，而我们的方法生成的部分几何与结构一致。对于第二个示例，我们的方法生成了几何特征协调的部分(b)，而使用解耦的结构和几何会导致部分风格不一致(a)。

边界框的分辨率。默认情况下，我们的方法使用每个包含19.2K个三角形的边界框。我们还尝试使用低分辨率和高分辨率的边界框。如图22所示，使用低分辨率 (b) 无法捕捉形状的细节，而使用高分辨率 (d) 产生的结果与我们的默认设置 (c) 非常相似，但需要更长时间。我们的默认设置 (c) 在效率和质量之间提供了良好的平衡。

单个部件的PartVAE与整体PartVAE对比。在我们的论文中，我们为每种部件类型训练一个PartVAE。我们将此方法与另一种替代方法进行比较，即对所有部件类别训练一个PartVAE。如图22 (e) 所示，这种方法在捕捉不同部件的独特几何特征方面效果不佳，导致几何重建质量较差。

泛化能力。图23展示了一个示例，证明了我们的方法在无需输入语义分割的情况下处理同类新形状的泛化能力。我们首先使用标记数据集训练PointNet++ [Qi等人 2017]，然后使用该数据集对新形状进行语义分割。最后，我们通过SDM-NET获得重建结果。图23展示了一个示例，证明了自动获得的语义分割结果可以有效地作为我们方法的输入。

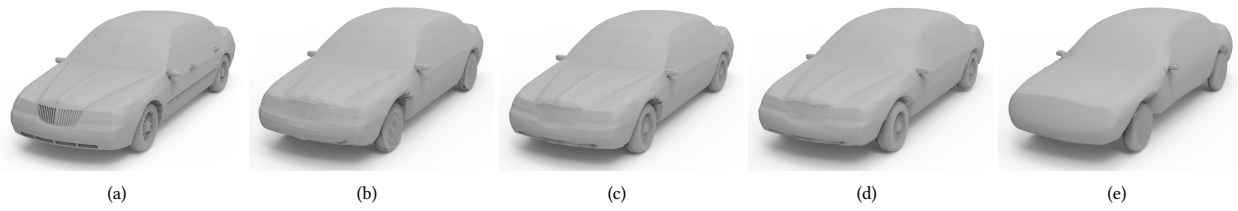


Fig. 22. Shape reconstruction using our SDM-NET with changing bounding box resolutions and using a single PartVAE for all part categories. (a) input shape, (b) our method with low-resolution bounding boxes (4.8K triangles), (c) our method with default resolution bounding boxes (19.2K triangles), (d) our method with high-resolution bounding boxes (76.8K triangles), (e) result with a single PartVAE for all part categories.

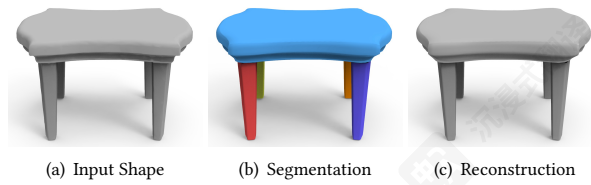


Fig. 23. An example demonstrating the generalizability of our method. (a) input shape without semantic segmentation, (b) segmentation by PointNet++, (c) the reconstruction result by our method.

Watertight models. The direct output of our method includes watertight meshes for individual parts, but not the shape as a whole. As demonstrated in Figure 24, by applying a watertight reconstruction technique [Huang et al. 2018], watertight meshes can be obtained, which benefit certain downstream applications.

Editability. Our generative model produces Structure Deformable Meshes, which are immediately editable in a structure-aware manner. This is difficult for other generative methods (e.g. [Li et al. 2017; Wu et al. 2016]). An example is given in Figure 25, which shows an editing sequence, including removing parts (when a part is removed, its symmetric part is also removed), making a chair leg longer, which also affects other chair legs due to the equal length constraint, and further deforming the chair back using an off-the-shelf deformation method [Sorkine and Alexa 2007]. During shape deformation, the editing constraints are treated as hard constraints and the equal length constraints are used in the refinement step (see Section 3.5).

Limitations. Although our method can handle a large variety of shapes with flexible structures and fine details, it still suffers from several limitations. While our method can handle shapes with holes formed by multiple parts, if a part itself has holes in it, our deformable box is unable to represent it exactly as the topology of parts cannot be different from genus-zero boxes. In this case, our method will try to preserve the mesh geometry but cannot maintain the hole. For certain parts which are unusual (e.g. the legs and back of the chair and the headstock of the guitar in Figure 26), our VAE architecture considers such cases as outliers, and “projects” them back to deformations consistent with the training set. Another limitation is that currently SDM-NET is trained using a collection of shapes with the same category. It thus cannot be used for interpolating shapes of different categories.

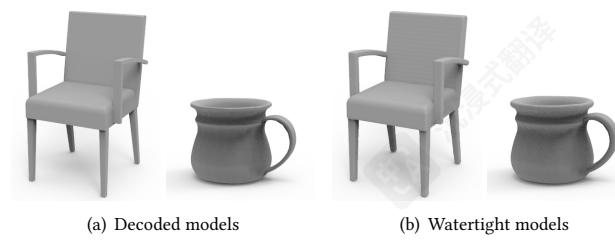


Fig. 24. Watertight models derived from our decoded models using [Huang et al. 2018]

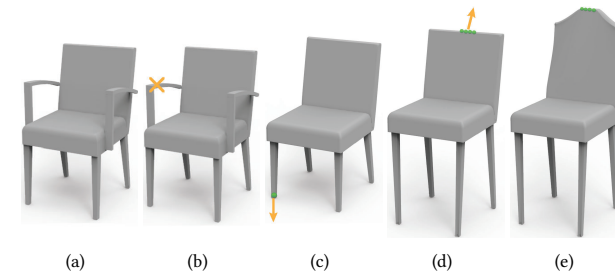


Fig. 25. Our generated Structured Deformable Mesh is directly editable. We show an example editing sequence. (a) is the decoded shape of our method. After applying the deletion operation (b), we obtain a chair without armrests (c). Dragging a leg of the chair makes all four legs longer due to the equal length constraint (d). Finally, we deform the back of the chair to obtain (e).

6 CONCLUSIONS AND FUTURE WORK

In this paper, we have presented SDM-NET, a novel deep generative model that generates 3D shapes as Structured Deformable Meshes. A shape is represented using a set of deformable boxes, and a two-level VAE is built to encode local geometry variations of individual parts, and global structure and geometries of all parts, respectively. Our representation achieves both flexible topology and fine geometric details, outperforming the state-of-the-art methods for both shape generation and shape interpolation.

As future work, our method could be generalized to reconstruct shapes from images. Similar to [Xin et al. 2018], which uses a deep neural network to learn the segmentation masks of cylinder regions from a given image for reconstructing 3D models composed

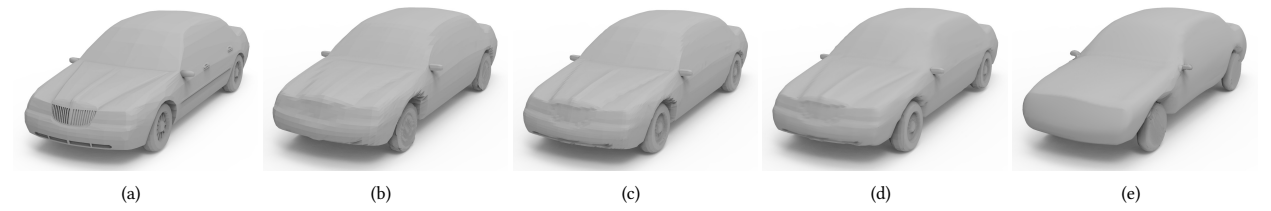


图22. 使用我们的SDM-NET进行形状重建，通过改变边界框分辨率并使用单个PartVAE处理所有部件类别。(a)输入形状，(b)使用低分辨率边界框（4.8K三角形）的我们的方法，(c)使用默认分辨率边界框（19.2K三角形）的我们的方法，(d)使用高分辨率边界框（76.8K三角形）的我们的方法，(e)对所有部件类别使用单个PartVAE的结果。



图23. 展示我们方法泛化能力的一个示例。(a) 无语义分割的输入形状，(b) PointNet++的分割结果，(c) 我们方法的重建结果。

水密模型。我们方法直接输出的单个部件包含水密网格，但整个形状并非如此。如图24所示，通过应用水密重建技术 [Huang等人 2018]，，可以获得水密网格，这有利于某些下游应用。

可编辑性。我们的生成模型产出结构可变形网格，这些网格能以结构感知的方式立即编辑。这对其他生成方法（例如 [Li 等人 2017; Wu 等人 2016]）来说很困难。图25给出了一个示例，展示了编辑序列，包括移除部分（当移除一部分时，其对称部分也会被移除）、将椅子腿变长（由于等长约束，这也会影响其他椅子腿），以及使用现成的变形方法 [Sorkine 和 Alexa 2007] 进一步变形椅子靠背。在形状变形过程中，编辑约束被视为硬约束，等长约束在精炼步骤中使用（见第3.5节）。

局限性。尽管我们的方法能够处理具有灵活结构和精细细节的各种形状，但它仍然存在一些局限性。虽然我们的方法可以处理由多个部件形成的带孔形状，但如果一个部件本身也有孔，我们的可变形框无法精确表示它，因为部件的拓扑结构与零类连通体不同。在这种情况下，我们的方法将尝试保留网格几何形状，但无法保持孔洞。对于某些不寻常的部件（例如图26中的椅腿和椅背以及吉他头），我们的VAE架构将此类情况视为异常值，并将它们“投影”回与训练集一致的变形。另一个局限性是，目前SDM-NET使用具有相同类别的形状集合进行训练。因此，它不能用于插值不同类别的形状。

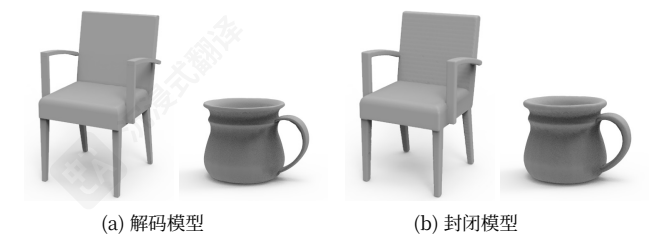


图.24. 使用 [Huang等人 2018]从我们的解码模型中推导出的封闭模型

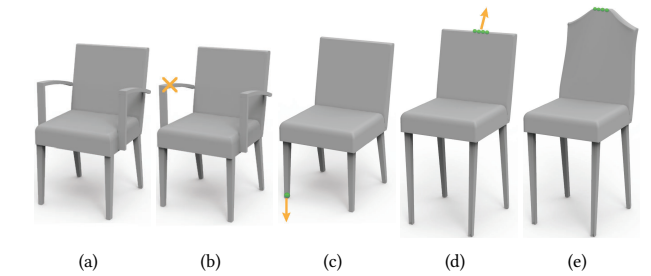


图25. 我们生成的结构化可变形网格可直接编辑。我们展示了一个编辑示例序列。(a) 是我们方法解码的形状。应用删除操作(b)后，我们得到一把没有扶手的椅子(c)。拖动椅子的腿使所有四条腿都变长，这是由于等长约束(d)。最后，我们变形椅子的背部，得到(e)。

6 结论与未来工作

在本文中，我们提出了SDM-NET，一种新型的深度生成模型，它将3D形状生成结构化可变形网格。形状使用一组可变形框表示，并构建了一个双层VAE，分别对单个部件的局部几何变化以及所有部件的全局结构和几何进行编码。我们的表示方法实现了灵活的拓扑结构和精细的几何细节，在形状生成和形状插值方面均优于现有技术。

作为未来的工作，我们的方法可以推广到从图像中重建形状。类似于 [Xin 等人 2018]，使用深度神经网络从给定图像中学习圆柱区域分割掩码以重建 3D 模型，该方法也适用于

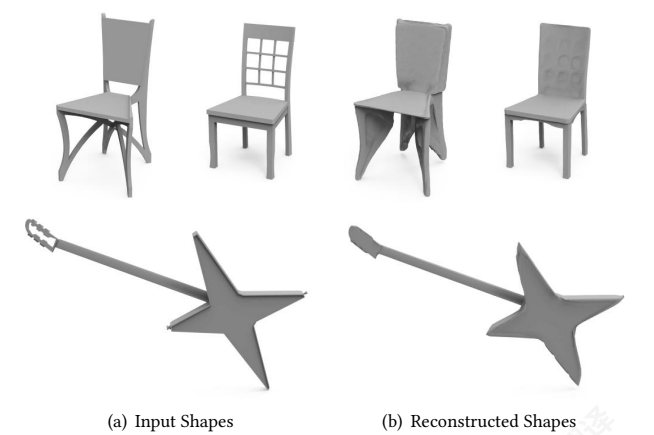


Fig. 26. Failure cases: the headstock of the guitar with a hole, the chair with complex legs and the chair with grid back could not be decoded by our SDM-NET since it requires each part to have the fixed topology, i.e., the same as a genus-zero box.

of cylindrical shapes, a possible approach to extend our method is to learn the segmentation of different parts in images and use such segmentation results as the conditions of the SP-VAE for 3D shape reconstruction. In this case, our SDM-NET makes it possible to generate rich shapes with details to better match given images. By exploiting the latent space of our network, our approach could also be generalized for data-driven deformation by incorporating user editing constraints in the optimization framework. It is also interesting to investigate how to extend our method to encode shapes of different categories using a single network. Our current approach can generate parts with the same resolution as the primitive bounding box mesh. We currently utilize a high-resolution mesh with fixed size, and therefore our generated shapes take up large storage space because of richer geometric details and higher resolution of meshes. However, since different kinds of parts have different geometric richness, it would be better to exploit (possibly different types of) primitives with adaptive resolutions for different parts so that we can preserve the same level of details but with significantly less storage space.

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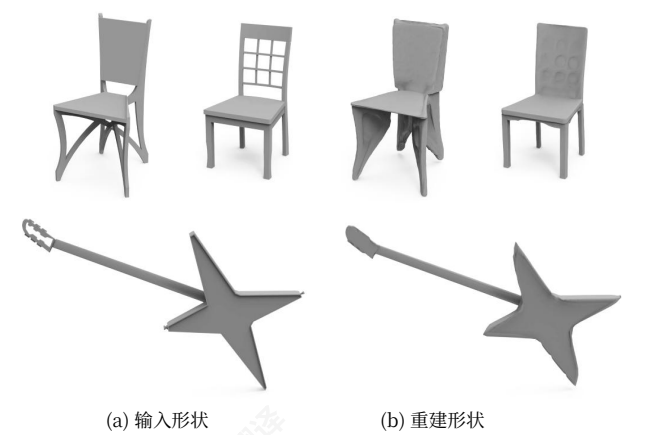


图26. 失效案例：带有孔的吉他琴头、腿复杂的椅子和带网格背面的椅子无法被我们的SDM-NET解码，因为它要求每个部分具有固定的拓扑结构，即与零亏格盒子相同。

我们当前的方法可以生成与原始边界框网格相同分辨率的部件。我们目前使用固定大小的高分辨率网格，因此由于更丰富的几何细节和更高的网格分辨率，生成的形状占用了大量的存储空间。然而，由于不同种类的部件具有不同的几何丰富性，使用具有自适应分辨率的（可能不同类型的）基元来处理不同部件会更好，这样我们可以在显著减少存储空间的同时保持相同级别的细节。

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APPENDIX: SUPPORT RELATIONSHIP FORMULATION.

Let $\tilde{b}_i$ and $\tilde{b}_j$ be the bounding boxes of parts i and j projected onto the plane orthogonal to the supporting direction. They should satisfy either $\tilde{b}_i \subseteq \tilde{b}_j$ or $\tilde{b}_j \subseteq \tilde{b}_i$. This constraint can be formulated as an integer programming problem and solved efficiently during the optimization as follows:

Let t_1 and t_2 be the two directions in the tangential plane. Denote by $\delta_1^{i,j}$ and $\delta_2^{i,j}$ two auxiliary binary variables, $\delta_1^{i,j}, \delta_2^{i,j} \in \{0, 1\}$, this is equivalent to

$$\begin{aligned} \mathbf{p}_j'[t_1] - \mathbf{q}_j'[t_1] &\leq \mathbf{p}_i'[t_1] - \mathbf{q}_i'[t_1] + M\delta_1^{i,j}, \\ \mathbf{p}_i'[t_1] + \mathbf{q}_i'[t_1] &\leq \mathbf{p}_j'[t_1] + \mathbf{q}_j'[t_1] + M\delta_1^{i,j}, \\ \mathbf{p}_j'[t_2] - \mathbf{q}_j'[t_2] &\leq \mathbf{p}_i'[t_2] - \mathbf{q}_i'[t_2] + M\delta_1^{i,j}, \\ \mathbf{p}_i'[t_2] + \mathbf{q}_i'[t_2] &\leq \mathbf{p}_j'[t_2] + \mathbf{q}_j'[t_2] + M\delta_1^{i,j}, \end{aligned} \quad (5)$$

$$\begin{aligned} \mathbf{p}_i'[t_1] - \mathbf{q}_i'[t_1] &\leq \mathbf{p}_j'[t_1] - \mathbf{q}_j'[t_1] + M\delta_2^{i,j}, \\ \mathbf{p}_j'[t_1] + \mathbf{q}_j'[t_1] &\leq \mathbf{p}_i'[t_1] + \mathbf{q}_i'[t_1] + M\delta_2^{i,j}, \\ \mathbf{p}_i'[t_2] - \mathbf{q}_i'[t_2] &\leq \mathbf{p}_j'[t_2] - \mathbf{q}_j'[t_2] + M\delta_2^{i,j}, \\ \mathbf{p}_j'[t_2] + \mathbf{q}_j'[t_2] &\leq \mathbf{p}_i'[t_2] + \mathbf{q}_i'[t_2] + M\delta_2^{i,j}, \end{aligned} \quad (6)$$

$$\delta_1^{i,j} + \delta_2^{i,j} \leq 1, \quad (7)$$

where M is a large positive number (larger than any possible coordinate in the shape), Eq. (7) is true if at most one of $\delta_1^{i,j}$ or $\delta_2^{i,j}$ can be 1, i.e., at least one of them is 0. Without loss of generality, assuming $\delta_1^{i,j} = 0$, then the set of equations in (5) without the term involving M is true, meaning $\tilde{b}_i \subseteq \tilde{b}_j$. Similarly, when $\delta_2^{i,j} = 0$, it satisfies that $\tilde{b}_j \subseteq \tilde{b}_i$.

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附录：支持关系公式化。

设 b_i 和 b_j 是部分 i 和 j 在垂直于支撑方向的正交平面上的边界框。它们应该满足 $b_i \subseteq b_j$ 或 $b_j \subseteq b_i$ 的条件。该约束可以表述为一个整数规划问题，并在优化过程中高效求解，具体如下：

设 t_1 和 t_2 为切平面上的两个方向。用 $\delta_1^{i,j}$ 和 $\delta_2^{i,j}$ 表示两个辅助二元变量， $\delta_1^{i,j}, \delta_2^{i,j} \in \{0, 1\}$ ，，则这相当于

$$\begin{aligned} \mathbf{p}_j'[t_1] - \mathbf{q}_j'[t_1] &\leq \mathbf{p}_i'[t_1] - \mathbf{q}_i'[t_1] + M\delta_1^{i,j}, \\ \mathbf{p}_i'[t_1] + \mathbf{q}_i'[t_1] &\leq \mathbf{p}_j'[t_1] + \mathbf{q}_j'[t_1] + M\delta_1^{i,j}, \\ \mathbf{p}_j'[t_2] - \mathbf{q}_j'[t_2] &\leq \mathbf{p}_i'[t_2] - \mathbf{q}_i'[t_2] + M\delta_1^{i,j}, \\ \mathbf{p}_i'[t_2] + \mathbf{q}_i'[t_2] &\leq \mathbf{p}_j'[t_2] + \mathbf{q}_j'[t_2] + M\delta_1^{i,j}, \end{aligned} \quad (5)$$

$$\begin{aligned} \mathbf{p}_i'[t_1] - \mathbf{q}_i'[t_1] &\leq \mathbf{p}_j'[t_1] - \mathbf{q}_j'[t_1] + M\delta_2^{i,j}, \\ \mathbf{p}_j'[t_1] + \mathbf{q}_j'[t_1] &\leq \mathbf{p}_i'[t_1] + \mathbf{q}_i'[t_1] + M\delta_2^{i,j}, \\ \mathbf{p}_i'[t_2] - \mathbf{q}_i'[t_2] &\leq \mathbf{p}_j'[t_2] - \mathbf{q}_j'[t_2] + M\delta_2^{i,j}, \\ \mathbf{p}_j'[t_2] + \mathbf{q}_j'[t_2] &\leq \mathbf{p}_i'[t_2] + \mathbf{q}_i'[t_2] + M\delta_2^{i,j}, \end{aligned} \quad (6)$$

$$\delta_1^{i,j} + \delta_2^{i,j} \leq 1, \quad (7)$$

其中 M 是一个很大的正数（大于形状中任何可能的坐标），如果 $\delta_{i,j}^1$ 或 $\delta_{i,j}^2$ 中至多有一个为 1，即至少有一个为 0，则等式 (7) 成立。不失一般性，假设 $\delta_1^{i,j} = 0$ ，，则 (5) 中的方程组在去掉项的情况下成立，这意味着 δ_1 。类似地，当 δ_2 涉及 M 的方程组也成立，即 $\tilde{b}_i \subseteq \tilde{b}_j$ 。